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The Korteweg-de Vries Equation With an Interface

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ABSTRACT

In this paper, we consider the Korteweg–de Vries (KdV) equation on the real line with an interface. Using Fokas's unified transform method, the explicit solution formulas for the linear forced KdV equation with an interface are derived. Building on these solution formulas, we establish standard estimates for the linear solution and a bilinear estimate for the nonlinear term in a suitable Sobolev space. Using these estimates and a contraction mapping argument, we prove the local well-posedness for the KdV equation with an interface.

1 | Introduction

The Korteweg–de Vries (KdV) equation

$$u_t + u_{xxx} + uu_x = 0, \tag{1.1}$$

is a nonlinear partial differential equation that models the unidirectional propagation of small-amplitude dispersive waves. Originally introduced by Boussinesq in 1877 to describe shallow water waves, it was later independently derived by Korteweg and de Vries in 1895. In 1965, the study of the KdV equation continued with the numerical experiments of Zabusky and Kruskal [1], which revealed the soliton interaction phenomenon. Soon after, Gardner, Greene, Kruskal, and Miura [2] solved the initial value problem for the KdV equation on the real line using the inverse scattering transform (IST) method. For further background on the KdV equation and its physical and mathematical significance, we refer readers to [3, 4]. For more detailed discussions of the IST method, see [5, 6] and the references therein.

On the other hand, significant attention has also been devoted to the well-posedness of the KdV equation. Relevant literature can be found in Bona and Smith [7], Saut and Temam [8], Constantin and Saut [9], Kenig, Ponce, and Vega [10, 11], Bourgain [12], and the references therein. Furthermore, a more general equation, namely, $u_t + u_{xxx} + u^m u_x = 0$, has also attracted significant attention. We refer the readers to Kenig, Ponce, and Vega [13], Grünrock [14], Colliander, Keel, Staffilani, Takaoka, and Tao [15], the monographs [16, 17], and the references therein.

Fokas developed a unified transform method, commonly known as the Fokas method, to solve initial-boundary value problems on domains such as the half-line and finite intervals [18]. This method can generate solution formulas for many cases that classical approaches cannot handle, particularly for equations involving higher order derivatives. Its generality is a significant advantage, broadening the method's applicability beyond the limitations of traditional techniques. Moreover, this method universally applies the same principles to Dirichlet, Neumann, and Robin boundary conditions, differing only in computational details. Consequently, it provides a unified and systematic approach to solving differential equations. For more details on the Fokas method, we refer readers to [19] and the monograph [20].

Another significant advantage of the Fokas method is that it enables one to determine which boundary conditions lead to a well-posed problem. This feature has been demonstrated in several equations, including the nonlinear Schrödinger equation [21], the KdV equation [22], the "good" Boussinesq equation [23], and the reaction–diffusion equation [24] in the half-line case, as well as for the KdV equation [25] and the reaction–diffusion equation [24] on finite intervals. Related studies have also addressed the KdV equation with different boundary conditions [26, 27] and higher order dispersive effects [28]. To the best of our knowledge, there are two other approaches to study the well-posedness of the initial boundary value problem for the KdV equation with Dirichlet boundary conditions. The first approach, introduced by Bona, Sun, and Zhang [29], utilizes the Laplace transform in the temporal variable to derive an explicit solution formula. The second approach, developed by Colliander and Kenig [30], recasts the initial boundary value problem as an initial value problem with a specially constructed forcing term. This forcing term is designed to satisfy the boundary condition by inverting a Riemann–Liouville fractional integral. Moreover, several developments on the half-line case of the generalized KdV equation have also been made; see Bona and Luo [31], Colliander and Kenig [30], Holmer [32], Compaan and Tzirakis [33], and the references therein.

Interface problems modeled by partial differential equations appear in many applications, such as computational electromagnetics [34], heat conduction between materials of different heat capacities [35–37], and incompressible two-phase flows [38]. Typically, these problems in partial differential equations lead to discontinuous or singular parameters across the interfaces in the solution domain. Thus, the solutions change dramatically across the interfaces. This poses significant challenges in both theoretical and numerical analysis (see [39]). Despite the Fokas method's original focus on boundary value problems, it has proven adaptable to certain interface problems in one dimension. Specifically, the Fokas method has been employed to derive solution formulas for interface problems involving linear dissipative and dispersive differential equations [40–46]. In particular, Deconinck, Sheils, and Smith [42] studied the following interface problem:

$$\begin{cases} u_t^L = \sigma_L^3 u_{xxx}^L, & x < 0,\ 0 < t \le T, \\ u_t^R = \sigma_R^3 u_{xxx}^R, & 0 < x,\ 0 < t \le T, \end{cases} \tag{1.2}$$

subject to the initial conditions

$$\begin{cases} u^L(x,0) = u_0^L(x), & x < 0, \\ u^R(x,0) = u_0^R(x), & x > 0, \end{cases} \tag{1.3}$$

with $u^L(\cdot,t) \in S((-\infty,0))$ and $u^R(\cdot,t) \in S((0,\infty))$ where $S(X)$ denotes the Schwartz space of restrictions to X of rapidly decaying functions. The interface conditions for (1.2) at $x = 0$ are expressed as an inhomogeneous linear combination of the terms $\frac{\partial^i}{\partial x^i} u^L(x,t)\Big|_{x=0}$ and $\frac{\partial^i}{\partial x^i} u^R(x,t)\Big|_{x=0}$ for $i = 0, 1, 2$.

Note that when employing the Fokas method to derive solutions for the linear problems (1.2), the number of required interface conditions depends on the signs of the constant functions σ_L and σ_R (see Section 5 in [42]). For example, in the case when $\sigma_L > 0$ and $\sigma_R < 0$, the interface conditions have the form

$$c_{11}u^R(0,t) + c_{12}u_x^R(0,t) + c_{13}u_{xx}^R(0,t) + c_{14}u^L(0,t) + c_{15}u_x^L(0,t) + c_{16}u_{xx}^L(0,t) = \mu_0(t), \tag{1.4a}$$

$$c_{21}u^R(0,t) + c_{22}u_x^R(0,t) + c_{23}u_{xx}^R(0,t) + c_{24}u^L(0,t) + c_{25}u_x^L(0,t) + c_{26}u_{xx}^L(0,t) = \mu_1(t), \tag{1.4b}$$

with given functions $\mu_0(t)$ and $\mu_1(t)$. The constants c_{ij} satisfy at least one of the following conditions:

1. $c_{11}c_{24} \ne c_{14}c_{21}$,
2. $\sigma_L\,(c_{12}c_{24} - c_{14}c_{22}) \ne \sigma_R\,(c_{15}c_{21} - c_{11}c_{25})$,

3. $\sigma_L^2(c_{13}c_{24} - c_{14}c_{23}) + \sigma_L\sigma_R(c_{12}c_{25} - c_{15}c_{22}) + \sigma_R^2(c_{11}c_{21} - c_{16}c_{21}) \neq 0$,
4. $\sigma_L(c_{13}c_{25} - c_{15}c_{23}) \neq \sigma_R(c_{16}c_{22} - c_{12}c_{26})$,
5. $c_{13}c_{26} \neq c_{16}c_{23}$.

In this paper, we continue the work of Deconinck, Sheils, and Smith [42] to study the local well-posedness problem for the KdV equation in a composite domain. Specifically, we consider the following equation:

$$\begin{cases} u_t^L = \sigma_L^3 u_{xxx}^L - u^L u_x^L, & -\infty < x < 0,\ 0 < t \leq T, \\ u_t^R = \sigma_R^3 u_{xxx}^R - u^R u_x^R, & 0 < x < \infty,\ 0 < t \leq T, \end{cases} \tag{1.5}$$

where $\sigma_L > 0$ and $\sigma_R < 0$. The solution $u(x,t)$ is defined by

$$u(x,t) = \begin{cases} u^L(x,t), & -\infty < x < 0,\ 0 \leq t \leq T, \\ u^R(x,t), & 0 < x < \infty,\ 0 \leq t \leq T. \end{cases} \tag{1.6}$$

We consider the initial data (1.3), as well as the following interface condition:

$$u^R(0,t) - u^L(0,t) = \mu_0(t), \qquad u_x^R(0,t) - u_x^L(0,t) = \mu_1(t). \tag{1.7}$$

Note that we focus on the special interface conditions (1.7) to avoid a lengthy discussion. It is straightforward to see that the coefficients in (1.4) are chosen as $c_{11} = c_{22} = 1$, $c_{14} = c_{25} = -1$, and all other $c_{ij} = 0$. This choice ensures that condition (2) is satisfied.

To introduce our main result, we shall define the fractional Sobolev space and solution space as follows:

Definition 1. Let $I = (a,b)$ with $-\infty \leq a < b \leq \infty$ and $0 < s < 1$. A function $f \in L^2(I)$ belongs to the fractional Sobolev space $H^s(I)$ if

$$\|f\|_{H^s(I)}^2 := \|f\|_{L^2(I)}^2 + [f]_s^2 < \infty, \tag{1.8}$$

where

$$[f]_s := \left(\int_I \int_I \frac{|f(x) - f(y)|^2}{|x-y|^{1+2s}}\, dxdy \right)^{\frac{1}{2}}. \tag{1.9}$$

Furthermore, for any positive number s, we write $s = \lfloor s \rfloor + \beta$, $0 \leq \beta < 1$. Then, the norm of the fractional Sobolev space can be expressed as

$$\|f\|_{H^s(I)}^2 = \sum_{j=0}^{\lfloor s \rfloor} \left\| \partial_x^j f \right\|_{L^2(I)}^2 + \left[\partial_x^{\lfloor s \rfloor} f \right]_\beta^2. \tag{1.10}$$

Definition 2. Let X_T be the space defined by $X_T := X_{-,T} + X_{+,T}$ and let

$$X_{-,T} := \left\{ u^L \in C([0,T]; H^s(\mathbb{R}^-)) : \|u^L\|_{X_{-,T}} < \infty \right\}, \quad X_{+,T} := \left\{ u^R \in C\big([0,T]; H^s(\mathbb{R}^+)\big) : \|u^R\|_{X_{+,T}} < \infty \right\}. \tag{1.11}$$

The norm for the space X_T is defined as $\|u\|_{X_T} := \|u^L\|_{X_{-,T}} + \|u^R\|_{X_{+,T}}$, where

$$\begin{aligned} \|u^L\|_{X_{-,T}} &:= \max\left\{ \Lambda_{-,1,s}^T(u^L), \Lambda_{-,2,s}^T(u^L), \Lambda_{-,3}^T(u^L), \Lambda_{-,4}^T(u^L) \right\},\ \|u^R\|_{X_{+,T}} \\ &:= \max\left\{ \Lambda_{+,1,s}^T(u^R), \Lambda_{+,2,s}^T(u^R), \Lambda_{+,3}^T(u^R), \Lambda_{+,4}^T(u^R) \right\}. \end{aligned} \tag{1.12}$$

The Λ-norms are denoted as follows:

$$\Lambda_{\pm,1,s}^T(u) := \sup_{t \in [0,T]} \|u(\cdot,t)\|_{H^s(\mathbb{R}\pm)}, \tag{1.13a}$$

$$\Lambda^T_{\pm,2,s}(u) := \sup_{x\in\mathbb{R}^\pm}\left(\int_0^T |D_x^s u_x(x,t)|^2\,dt\right)^{\frac{1}{2}}, \tag{1.13b}$$

$$\Lambda^T_{\pm,3}(u) := \left(\int_{\mathbb{R}^\pm}\sup_{t\in[0,T]} |u(x,t)|^2\,dx\right)^{\frac{1}{2}}, \tag{1.13c}$$

$$\Lambda^T_{\pm,4}(u) := \left(\int_0^T \sup_{x\in\mathbb{R}^\pm} |u_x(x,t)|^4\,dt\right)^{\frac{1}{4}}, \tag{1.13d}$$

where the subscripts + and − indicate the regions of x. Specifically, $\Lambda^T_{-,1,s}(u) = \Lambda^T_{-,1,s}(u^L)$ and $\Lambda^T_{+,1,s}(u) = \Lambda^T_{+,1,s}(u^R)$, with the other norms following the same pattern. In addition, $\Lambda^T_{\pm,2,s}(u)$ can be expressed as

$$\sup_{x\in\mathbb{R}^\pm}\left(\int_0^T |D_x^s u_x(x,t)|^2\,dt\right)^{\frac{1}{2}} = \sup_{x\in\mathbb{R}^\pm}\left(\int_0^T\left(\int_{\mathbb{R}^\pm}\frac{|u_x(x+z,t)-u_x(x,t)|^2}{|z|^{1+2s}}dz\right)dt\right)^{\frac{1}{2}}. \tag{1.14}$$

Theorem 1.1. *Given $\sigma_L > 0$, $\sigma_R < 0$. Let $u_0^L \in H^s(\mathbb{R}^-)$, $u_0^R \in H^s(\mathbb{R}^+)$, $\mu_0 \in H^{(s+1)/3}((0,T))$, and $\mu_1 \in H^{s/3}((0,T))$. Assume $3/4 < s < 1$, $0 < T \le 1$ and the initial condition $u_0^L(x)$ and $u_0^R(x)$ satisfy the following interface conditions:*

$$u_0^L(0) = u_0^R(0), \qquad (u_0^L)'(0) = (u_0^R)'(0), \tag{1.15}$$

then there exist a short time $0 < T^* \le T$ and a unique solution $u \in X_{T^*}$ to Equation (1.5) with the initial data (1.3) and the interface conditions (1.7).

To study the existence of the KdV equation with an interface, we first consider the following linear forced equations:

$$\begin{cases} v_t^L = \sigma_L^3 v_{xxx}^L + f_L(x,t), & -\infty < x < 0,\ 0 < t \le T, \\ v_t^R = \sigma_R^3 v_{xxx}^R + f_R(x,t), & 0 < x < \infty,\ 0 < t \le T, \end{cases} \tag{1.16}$$

with the initial data (1.3), interface conditions (1.7), and the given forcing term $f(x,t)$, where $f_L(x,t) = f(x,t)\mathbb{1}_{(-\infty,0)}(x)$ and $f_R(x,t) = f(x,t)\mathbb{1}_{(0,\infty)}(x)$ for $0 < t \le T$. By employing the Fokas method, the solutions to the linear equations (1.16) can be formulated as follows:

$$\begin{aligned}
v^L(x,t) &= S_L[u_0,\mu_0,\mu_1;f] \\
&:= \frac{1}{2\pi}\int_{\mathbb{R}} e^{ikx-ik^3\sigma_L^3 t}\left[\widehat{u_0^L}(k) + F_L(k,t)\right]dk \\
&\quad + \frac{\sigma_L+\alpha\sigma_L+\sigma_R}{2\pi\alpha\sigma_L(\sigma_L-\sigma_R)}\int_{\partial D_5} e^{i\frac{k}{\sigma_L}x-ik^3t}\left[\widehat{u_0^L}\left(\frac{\alpha k}{\sigma_L}\right) + F_L\left(\frac{\alpha k}{\sigma_L},t\right)\right]dk \\
&\quad - \frac{\sigma_L+\sigma_R+\alpha\sigma_R}{2\pi\alpha\sigma_L(\sigma_L-\sigma_R)}\int_{\partial D_5} e^{i\frac{k}{\sigma_L}x-ik^3t}\left[\widehat{u_0^L}\left(\frac{\alpha^2 k}{\sigma_L}\right) + F_L\left(\frac{\alpha^2 k}{\sigma_L},t\right)\right]dk \\
&\quad + \frac{\sigma_L(2+\alpha)}{2\pi\alpha\sigma_R(\sigma_L-\sigma_R)}\int_{\partial D_5} e^{i\frac{k}{\sigma_L}x-ik^3t}\left[\widehat{u_0^R}\left(\frac{\alpha k}{\sigma_R}\right) + F_R\left(\frac{\alpha k}{\sigma_R},t\right)\right]dk \\
&\quad - \frac{\sigma_L(2+\alpha)}{2\pi\alpha\sigma_R(\sigma_L-\sigma_R)}\int_{\partial D_5} e^{i\frac{k}{\sigma_L}x-ik^3t}\left[\widehat{u_0^R}\left(\frac{\alpha^2 k}{\sigma_R}\right) + F_R\left(\frac{\alpha^2 k}{\sigma_R},t\right)\right]dk \\
&\quad - \frac{(2-\alpha-\alpha^2)\sigma_L}{2\pi(\sigma_L-\sigma_R)}\int_{\partial D_5} e^{i\frac{k}{\sigma_L}x-ik^3t}\left[ik\sigma_R M_1(ik^3,T) + k^2 M_0(ik^3,T)\right]dk,
\end{aligned} \tag{1.17a}$$

$$
\begin{aligned}
v^R(x,t) &= S_R[u_0,\mu_0,\mu_1;f] \\
&:= \frac{1}{2\pi}\int_{\mathbb{R}} e^{ikx-ik^3\sigma_R^3 t}\left[\widehat{u_0^R}(k)+F_R(k,t)\right]dk \\
&\quad - \frac{(2+\alpha)\sigma_R}{2\pi\alpha\sigma_L(\sigma_L-\sigma_R)}\int_{\partial D_5} e^{i\frac{k}{\sigma_R}x-ik^3t}\left[\widehat{u_0^L}\left(\frac{\alpha k}{\sigma_L}\right)+F_L\left(\frac{\alpha k}{\sigma_L},t\right)\right]dk \\
&\quad + \frac{(2+\alpha)\sigma_R}{2\pi\alpha\sigma_L(\sigma_L-\sigma_R)}\int_{\partial D_5} e^{i\frac{k}{\sigma_R}x-ik^3t}\left[\widehat{u_0^L}\left(\frac{\alpha^2 k}{\sigma_L}\right)+F_L\left(\frac{\alpha^2 k}{\sigma_L},t\right)\right]dk \\
&\quad - \frac{\sigma_L+\sigma_R+\alpha\sigma_R}{2\pi\alpha\sigma_R(\sigma_L-\sigma_R)}\int_{\partial D_5} e^{i\frac{k}{\sigma_R}x-ik^3t}\left[\widehat{u_0^R}\left(\frac{\alpha k}{\sigma_R}\right)+F_R\left(\frac{\alpha k}{\sigma_R},t\right)\right]dk \\
&\quad + \frac{\sigma_L+\alpha\sigma_L+\sigma_R}{2\pi\alpha\sigma_R(\sigma_L-\sigma_R)}\int_{\partial D_5} e^{i\frac{k}{\sigma_R}x-ik^3t}\left[\widehat{u_0^R}\left(\frac{\alpha^2 k}{\sigma_R}\right)+F_R\left(\frac{\alpha^2 k}{\sigma_R},t\right)\right]dk \\
&\quad + \frac{(2-\alpha-\alpha^2)\sigma_R}{2\pi(\sigma_L-\sigma_R)}\int_{\partial D_5} e^{i\frac{k}{\sigma_R}x-ik^3t}\left[ik\sigma_L M_1(ik^3,T)+k^2M_0(ik^3,T)\right]dk,
\end{aligned}
\tag{1.17b}
$$

where

$$
D_5 := \left\{k\in\mathbb{C} : k=|k|e^{i\theta},\ \frac{4\pi}{3}<\theta<\frac{5\pi}{3}\right\},\quad \alpha := e^{i\frac{2\pi}{3}}, \tag{1.18a}
$$

$$
\widehat{u_0^L}(k) := \int_{-\infty}^0 e^{-ikx}u_0^L(x)\,dx,\quad \widehat{u_0^R}(k) := \int_0^\infty e^{-ikx}u_0^R(x)\,dx, \tag{1.18b}
$$

$$
F_L(k,t) := \int_0^t e^{ik^3\sigma_L^3 t'}\widehat{f_L}(k,t')\,dt',\quad F_R(k,t) := \int_0^t e^{ik^3\sigma_R^3 t'}\widehat{f_R}(k,t')\,dt', \tag{1.18c}
$$

$$
M_0(ik^3,t) := \int_0^t e^{ik^3t'}\mu_0(t')\,dt',\quad M_1(ik^3,t) := \int_0^t e^{ik^3t'}\mu_1(t')\,dt'. \tag{1.18d}
$$

Remark 1.2. If the signs of the constants σ_L and σ_R change, the integrals along the path ∂D_5 will be altered. Indeed, in the case when $\sigma_L<0$ and $\sigma_R>0$, the paths in the integrals $v^L(x,t)$ and $v^R(x,t)$ both become $\partial D_1\cup\partial D_3$, where

$$
D_1 := \left\{k\in\mathbb{C} : k=|k|e^{i\theta},\ 0<\theta<\frac{\pi}{3}\right\},\quad D_3 := \left\{k\in\mathbb{C} : k=|k|e^{i\theta},\ \frac{2\pi}{3}<\theta<\pi\right\}. \tag{1.19}
$$

In the case when $\sigma_L>0$ and $\sigma_R>0$, the path in integral $v^L(x,t)$ does not change, while the path in integral $v^R(x,t)$ becomes $\partial D_1\cup\partial D_3$. Note that the case of $\sigma_L<0$ and $\sigma_R<0$ could be considered by letting x to $-x$ in the case when $\sigma_L>0$ and $\sigma_R>0$ (see [42] for more details).

Furthermore, we have the following linear estimates. The detail of the proof is postponed to Section 3.

Theorem 1.3. *Let $0<T\le 1$. Assume that $u_0^L\in H^s(\mathbb{R}^-)$, $u_0^R\in H^s(\mathbb{R}^+)$, $\mu_0\in H^{(s+1)/3}((0,T))$, and $\mu_1\in H^{s/3}((0,T))$. Then, the solutions (1.17a) and (1.17b) to Equation (1.16) with the initial data (1.3) and the interface conditions (1.7), (1.15) satisfy the following estimate:*

$$
\sup_{t\in[0,T]}\|v^L(\cdot,t)\|_{H^s(\mathbb{R}^-)}+\sup_{t\in[0,T]}\|v^R(\cdot,t)\|_{H^s(\mathbb{R}^+)} \lesssim_{s,\sigma_L,\sigma_R} \|(u_0,\mu_0,\mu_1)\|_D+\int_0^T\Big(\|f_L(\cdot,t)\|_{H^s(\mathbb{R}^-)}+\|f_R(\cdot,t)\|_{H^s(\mathbb{R}^+)}\Big)dt, \tag{1.20}
$$

where $1/2 < s < 1$ *and*

$$\|(u_0, \mu_0, \mu_1)\|_D := \|u_0^L\|_{H^s(\mathbb{R}^-)} + \|u_0^R\|_{H^s(\mathbb{R}^+)} + \|\mu_0\|_{H^{\frac{s+1}{3}}((0,T))} + \|\mu_1\|_{H^{\frac{s}{3}}((0,T))}. \tag{1.21}$$

Moreover, we have the following estimates to make sure that the linear solutions $v^L(x,t)$ *and* $v^R(x,t)$ *satisfy the interface conditions (1.7) in the low regularity case.*

$$\sup_{x\in\mathbb{R}^-} \|v^L(x,\cdot)\|_{H^{\frac{s+1}{3}}((0,T))} + \sup_{x\in\mathbb{R}^+} \|v^R(x,\cdot)\|_{H^{\frac{s+1}{3}}((0,T))} \lesssim_{s,\sigma_L,\sigma_R} \|(u_0,\mu_0,\mu_1)\|_D + \|f_L\|_{L^2([0,T];H^s(\mathbb{R}^-))} + \|f_R\|_{L^2([0,T];H^s(\mathbb{R}^+))}, \tag{1.22a}$$

and

$$\sup_{x\in\mathbb{R}^-} \|v_x^L(x,\cdot)\|_{H^{\frac{s}{3}}((0,T))} + \sup_{x\in\mathbb{R}^+} \|v_x^R(x,\cdot)\|_{H^{\frac{s}{3}}((0,T))} \lesssim_{s,\sigma_L,\sigma_R} \|(u_0,\mu_0,\mu_1)\|_D + \|f_L\|_{L^2([0,T];H^s(\mathbb{R}^-))} + \|f_R\|_{L^2([0,T];H^s(\mathbb{R}^+))}, \tag{1.22b}$$

where $1/2 < s < 3/2$.

To prove Theorem 1.1 (see Section 2), we need to handle the estimate for the nonlinear term. For this purpose, we recall the following lemma established in [22].

Lemma A (Lemma 3.1 in [22]). *Let* $0 \le s < 1$ *and* $0 < T \le 1$. *Then, we have*

$$\int_0^T \|uv_x\|_{H^s(\mathbb{R}^+)} dt \le C \max\{T^{\frac{1}{2}}, T^{\frac{3}{4}}\} \|u\|_{X_{+,T}} \|v\|_{X_{+,T}}, \tag{1.23}$$

where C *is a constant.*

As a consequence of Lemma A, we obtain the following Corollary (see Section 4).

Corollary 1.4. *Let* $0 \le s < 1$ *and* $0 < T \le 1$. *Then, we have*

$$\int_0^T \|uu_x\|_{H^s(\mathbb{R}^\pm)} dt \le C \max\{T^{\frac{1}{2}}, T^{\frac{3}{4}}\} \|u\|^2_{X_{\pm,T}}, \tag{1.24}$$

where C *is a constant.*

Finally, to complete the well-posedness results, we define the map $\Phi : X_T \to X_T$ as follows:

$$\Phi u := S_L[u_0, \mu_0, \mu_1; -uu_x] + S_R[u_0, \mu_0, \mu_1; -uu_x]. \tag{1.25}$$

The following theorem establishes the local Lipschitz continuity of the mapping Φ.

Theorem 1.5. *Suppose that we are given two distinct data* (u_0, μ_0, μ_1) *and* $(\tilde{u}_0, \tilde{\mu}_0, \tilde{\mu}_1)$ *that satisfy the same conditions stated in Theorem 1.1. Then, for* $3/4 < s < 1$, *there exists a time* $0 < T_c \le T$ *such that* u *and* $\tilde{u}$ *are fixed points of the maps* Φu *and* $\tilde{\Phi}\tilde{u}$, *respectively. Furthermore, there exists* $r_c > 0$ *such that if* $\|u\|_{X_{T_c}}, \|\tilde{u}\|_{X_{T_c}} \le r_c$, *then*

$$\|u - \tilde{u}\|_{X_{T_c}} = \|\Phi u - \tilde{\Phi}\tilde{u}\|_{X_{T_c}} \le 2C(s, \sigma_L, \sigma_R)\|(u_0, \mu_0, \mu_1) - (\tilde{u}_0, \tilde{\mu}_0, \tilde{\mu}_1)\|_D, \tag{1.26}$$

where C *is a constant depending on* s, σ_L, σ_R.

This paper presents a result for studying the KdV equation with an interface, specifically proving local well-posedness in the cases when $3/4 < s < 1$. The well-posedness of the KdV equation has been extensively studied on both the whole line and the half-line in low-regularity settings. In particular, local well-posedness has been established for $s > -3/4$ using Sobolev and Bourgain spaces. Meanwhile, for more general nonlinearities of the form $u^m u_x$ with $m \ge 2$, related local well-posedness results have been obtained, such as $s_2 \geq 1/4$, $s_3 \geq -1/6$, and $s_m \geq (m-4)/2m$ for $m \geq 4$. Here, s_m refers to the Sobolev exponent corresponding to the nonlinear term $u^m u_x$ with a given power m. The present work provides a first step in studying the interface problem for the KdV equation, and we observe that a gap remains when compared to the well-established results for the whole-line and half-line cases. We shall study these topics further in a forthcoming project. As a related direction, we also note that several studies have investigated the KdV equation with a linear transport term on the half-line. Motivated by this, we also aim to explore the corresponding problem on a composite domain in future work.

The paper is organized as follows. In Section 2, we establish Theorems 1.1 and 1.5. In Section 3, we derive standard linear estimates (1.20) and (1.22) for the solutions of linear forced equations (1.16). In Section 4, we demonstrate the solution space X_T using the bilinear estimates. To invoke the contraction mapping theorem, we prove the Λ-norm estimates of the linear solutions in Section 5.

2 | Solving the Interface Problem of KdV Equation

In this section, we establish the well-posedness of the interface problem for the equation (1.5). Specifically, we first establish the existence and uniqueness of the solution in Subsection 2.1. Then, in Subsection 2.2, we demonstrate that the iteration map Φ, defined in (1.25), is locally Lipschitz continuous in the solution space.

2.1 | Proof of Theorem 1.1

For a fixed time $0 < T^* \leq T \leq 1$, we let

$$B(\rho) := \{u \in X_{T^*} : \|u\|_{X_{T^*}} \leq \rho\}, \qquad \rho := 2C(s,\sigma_L,\sigma_R)\|(u_0,\mu_0,\mu_1)\|_D, \tag{2.1}$$

where $\|(u_0,\mu_0,\mu_1)\|_D$ is defined in (1.21). To show that the map $\Phi : B(\rho) \to B(\rho)$, we rely on the following estimates, whose proofs are postponed to Section 5.

Theorem 2.1. *Let $3/4 < s < 1$ and $0 < T \leq 1$. Assume that $u_0^L \in H^s(\mathbb{R}^-)$, $u_0^R \in H^s(\mathbb{R}^+)$, $\mu_0 \in H^{(s+1)/3}((0,T))$, and $\mu_1 \in H^{s/3}((0,T))$. Then, the solutions (1.17a) and (1.17b) of Equation (1.16) with the initial data (1.3) and interface conditions (1.7), (1.15) satisfy the following estimates:*

$$\|S_L[u_0,\mu_0,\mu_1;f]\|_{X_{-,T}} \lesssim_{s,\sigma_L,\sigma_R} \|(u_0,\mu_0,\mu_1)\|_D + \int_0^T \left(\|f_L(\cdot,t)\|_{H^s(\mathbb{R}^-)} + \|f_R(\cdot,t)\|_{H^s(\mathbb{R}^+)}\right) dt, \tag{2.2a}$$

$$\|S_R[u_0,\mu_0,\mu_1;f]\|_{X_{+,T}} \lesssim_{s,\sigma_L,\sigma_R} \|(u_0,\mu_0,\mu_1)\|_D + \int_0^T \left(\|f_L(\cdot,t)\|_{H^s(\mathbb{R}^-)} + \|f_R(\cdot,t)\|_{H^s(\mathbb{R}^+)}\right) dt. \tag{2.2b}$$

The notation $f \lesssim g$ indicates that there is a constant C such that $f \leq Cg$.

Using Theorem 2.1 and Corollary 1.4, we obtain

$$\begin{aligned}
\|\Phi u\|_{X_{T^*}} &= \|S_L[u_0,\mu_0,\mu_1;-uu_x]\|_{X_{-,T^*}} + \|S_R[u_0,\mu_0,\mu_1;-uu_x]\|_{X_{+,T^*}} \\
&\leq C(s,\sigma_L,\sigma_R)\left(\|(u_0,\mu_0,\mu_1)\|_D + \int_0^{T^*} \left(\|uu_x\|_{H^s(\mathbb{R}^-)} + \|uu_x\|_{H^s(\mathbb{R}^+)}\right) dt\right) \\
&\leq C(s,\sigma_L,\sigma_R)\left(\|(u_0,\mu_0,\mu_1)\|_D + 2(T^*)^{\frac{1}{2}}\|u\|_{X_{T^*}}^2\right).
\end{aligned} \tag{2.3}$$

Furthermore, if $u \in B(\rho)$ and

$$T^* \leq \frac{1}{256\, C(s,\sigma_L,\sigma_R)^4\, \|(u_0,\mu_0,\mu_1)\|_D^2}, \tag{2.4}$$

then

$$\|\Phi u\|_{X_{T^*}} \leq \frac{\rho}{2} + 2C(s,\sigma_L,\sigma_R)(T^*)^{\frac{1}{2}}\rho^2 \leq \rho. \tag{2.5}$$

The map Φ is a contraction map. Let $u_1, u_2 \in B(\rho)$ be two solutions with the same given data. By applying Theorem 2.1, we obtain

$$\begin{aligned}\|\Phi u_1 - \Phi u_2\|_{X_{T^*}} &= \|S_L[0,0,0;u_2(u_2)_x - u_1(u_1)_x]\|_{X_{-,T^*}} + \|S_R[0,0,0;u_2(u_2)_x - u_1(u_1)_x]\|_{X_{+,T^*}} \\ &\leq C(s,\sigma_L,\sigma_R)\int_0^{T^*}\left(\|u_2(u_2)_x - u_1(u_1)_x\|_{H^s(\mathbb{R}^-)} + \|u_2(u_2)_x - u_1(u_1)_x\|_{H^s(\mathbb{R}^+)}\right)dt. \end{aligned} \tag{2.6}$$

Note that

$$u_2(u_2)_x - u_1(u_1)_x = \frac{1}{2}[(u_1+u_2)_x(u_2-u_1) + (u_1+u_2)(u_2-u_1)_x]. \tag{2.7}$$

Combining (2.6), (2.7), and Corollary 1.4, we derive the following inequality:

$$\begin{aligned}\|\Phi u_1 - \Phi u_2\|_{X_{T^*}} &\leq 2C(s,\sigma_L,\sigma_R)(T^*)^{\frac{1}{2}}\left(\|u_1\|_{X_{T^*}} + \|u_2\|_{X_{T^*}}\right)\|u_1-u_2\|_{X_{T^*}} \\ &\leq 4C(s,\sigma_L,\sigma_R)(T^*)^{\frac{1}{2}}\rho\|u_1-u_2\|_{X_{T^*}} \\ &\leq \frac{1}{2}\|u_1-u_2\|_{X_{T^*}}, \end{aligned} \tag{2.8}$$

where the condition (2.4) is used. Therefore, choosing

$$T^* = \min\left\{T, \frac{1}{256\,C(s,\sigma_L,\sigma_R)^4\|(u_0,\mu_0,\mu_1)\|_D^2}\right\}, \tag{2.9}$$

the map Φ is a contraction map.

2.2 | Proof of Theorem 1.5

To complete the proof of well-posedness for the KdV equation with an interface, it remains to show that the data-to-solution map is locally Lipschitz continuous. Let (u_0,μ_0,μ_1) and $(\tilde{u}_0,\tilde{\mu}_0,\tilde{\mu}_1)$ be two data in a ball $\mathscr{B}(\ell) \subset H^s(\mathbb{R}) \times H^{(s+1)/3}((0,T)) \times H^{s/3}((0,T))$ with radius $\ell > 0$, centered at the origin. As we proved in Theorem 1.1, the lifespans (2.9) of the solutions u and $\tilde{u}$, denoted by T_u and $T_{\tilde{u}}$, are defined by

$$T_u := \min\left\{T, \frac{1}{256\,C(s,\sigma_L,\sigma_R)^4\|(u_0,\mu_0,\mu_1)\|_D^2}\right\}, \quad T_{\tilde{u}} := \min\left\{T, \frac{1}{256\,C(s,\sigma_L,\sigma_R)^4\|(\tilde{u}_0,\tilde{\mu}_0,\tilde{\mu}_1)\|_D^2}\right\}. \tag{2.10}$$

Since $\max\left\{\|(u_0,\mu_0,\mu_1)\|_D, \|(\tilde{u}_0,\tilde{\mu}_0,\tilde{\mu}_1)\|_D\right\} \leq \ell$, we may take

$$T_c := \min\left\{T, \frac{1}{256\,C(s,\sigma_L,\sigma_R)^4\ell^2}\right\} \leq \min\{T_u, T_{\tilde{u}}\}. \tag{2.11}$$

For any two solutions $u, \tilde{u} \in B(r_c) \subset X_{T_c}$ with data in $\mathscr{B}(\ell)$, we will determine the radius r_c such that the inequality (1.26) holds. By using the inequalities (2.6) and (2.7) and Theorem 2.1, as well as the inequality in Corollary 1.4, we deduce

$$\|u-\tilde{u}\|_{X_{T_c}} = \|\Phi u - \tilde{\Phi}\tilde{u}\|_{X_{T_c}} \leq C(s,\sigma_L,\sigma_R)\left(\|(u_0-\tilde{u}_0,\mu_0-\tilde{\mu}_0,\mu_1-\tilde{\mu}_1)\|_D + 4(T_c)^{\frac{1}{2}}r_c\|u-\tilde{u}\|_{X_{T_c}}\right). \tag{2.12}$$

By rearranging the inequality, we obtain

$$\|u-\tilde{u}\|_{X_{T_c}} \leq \frac{C(s,\sigma_L,\sigma_R)}{1-4C(s,\sigma_L,\sigma_R)(T_c)^{\frac{1}{2}} r_c} \|(u_0-\tilde{u}_0, \mu_0-\tilde{\mu}_0, \mu_1-\tilde{\mu}_1)\|_D, \tag{2.13}$$

provided that $1-4C(s,\sigma_L,\sigma_R)(T_c)^{1/2}r_c > 0$. Therefore, if we choose

$$r_c = \frac{1}{8\,C(s,\sigma_L,\sigma_R)T_c^{\frac{1}{2}}}, \tag{2.14}$$

then the estimate (1.26) holds.

3 | Linear Estimates of the Linear Forced Solutions

In this section, we prove the estimates (1.20) and (1.22) with the precise statements presented in Theorem 1.3. To establish these estimates, we first split the solution formulas (1.17a) and (1.17b) into several components. Let

$$v^L(x,t) = \mathcal{U}_L(x,t) + \mathcal{F}_L(x,t) + \sum_{j=1}^{4} C_j(\sigma_L,\sigma_R)\Big(v_j^-(x,t) + w_j^-(x,t)\Big) + C_0(\sigma_L,\sigma_R)\big(H_0^-(x,t) + \sigma_R H_1^-(x,t)\big) \tag{3.1a}$$

and

$$v^R(x,t) = \mathcal{U}_R(x,t) + \mathcal{F}_R(x,t) + \sum_{j=5}^{8} \tilde{C}_j(\sigma_L,\sigma_R)\Big(v_j^+(x,t) + w_j^+(x,t)\Big) + \tilde{C}_0(\sigma_L,\sigma_R)\big(H_0^+(x,t) + \sigma_L H_1^+(x,t)\big), \tag{3.1b}$$

where $C_j(\sigma_L,\sigma_R)$ and $\tilde{C}_j(\sigma_L,\sigma_R)$ are constants depending on σ_L and σ_R. Also, the functions $\mathcal{U}_{(\cdot)}(x,t)$, $\mathcal{F}_{(\cdot)}(x,t)$, $v_j^\pm(x,t)$, $w_j^\pm(x,t)$, and $H_i^\pm(x,t)$ are defined below.

$$\mathcal{U}_L(x,t) := \frac{1}{2\pi}\int_{\mathbb{R}} e^{ikx-ik^3\sigma_L^3 t}\widehat{u_0^L}(k)\,dk, \quad \mathcal{U}_R(x,t) := \frac{1}{2\pi}\int_{\mathbb{R}} e^{ikx-ik^3\sigma_R^3 t}\widehat{u_0^R}(k)\,dk, \tag{3.2a}$$

$$\mathcal{F}_L(x,t) := \frac{1}{2\pi}\int_{\mathbb{R}} e^{ikx-ik^3\sigma_L^3 t}F_L(k,t)\,dk, \quad \mathcal{F}_R(x,t) := \frac{1}{2\pi}\int_{\mathbb{R}} e^{ikx-ik^3\sigma_R^3 t}F_R(k,t)\,dk. \tag{3.2b}$$

The functions $v_j^\pm(x,t)$ are denoted by

$$v_j^-(x,t) := \int_{\partial D_5} e^{i\frac{k}{\sigma_L}x-ik^3t}\,\square_j\,dk, \qquad \text{for } x<0, \qquad j=1,2,3,4, \tag{3.3a}$$

$$v_j^+(x,t) := \int_{\partial D_5} e^{i\frac{k}{\sigma_R}x-ik^3t}\,\square_j\,dk, \qquad \text{for } x>0, \qquad j=5,6,7,8, \tag{3.3b}$$

where the superscripts + and − are used to indicate the region of x and

$$\square_1 = \square_5 = \widehat{u_0^L}\left(\frac{\alpha k}{\sigma_L}\right), \square_2 = \square_6 = \widehat{u_0^L}\left(\frac{\alpha^2 k}{\sigma_L}\right), \square_3 = \square_7 = \widehat{u_0^R}\left(\frac{\alpha k}{\sigma_R}\right), \square_4 = \square_8 = \widehat{u_0^R}\left(\frac{\alpha^2 k}{\sigma_R}\right). \tag{3.3c}$$

The functions $w_j^\pm(x,t)$ are denoted by

$$w_j^-(x,t) := \int_{\partial D_5} e^{i\frac{k}{\sigma_L}x-ik^3t}\,\tilde{\square}_j\,dk, \qquad \text{for } x<0, \qquad j=1,2,3,4, \tag{3.4a}$$

$$w_j^+(x,t) := \int_{\partial D_5} e^{i\frac{k}{\sigma_R}x-ik^3t}\,\tilde{\square}_j\,dk, \qquad \text{for } x>0, \qquad j=5,6,7,8, \tag{3.4b}$$

where

$$\tilde{\Box}_1=\tilde{\Box}_5=F_L\left(\frac{\alpha k}{\sigma_L},t\right),\ \tilde{\Box}_2=\tilde{\Box}_6=F_L\left(\frac{\alpha^2 k}{\sigma_L},t\right),\ \tilde{\Box}_3=\tilde{\Box}_7=F_R\left(\frac{\alpha k}{\sigma_R},t\right),\ \tilde{\Box}_4=\tilde{\Box}_8=F_R\left(\frac{\alpha^2 k}{\sigma_R},t\right). \tag{3.4c}$$

The functions $H_0^{\pm}(x,t)$ and $H_1^{\pm}(x,t)$ are denoted by

$$H_0^-(x,t):=\int_{\partial D_5} e^{i\frac{k}{\sigma_L}x-ik^3t}k^2M_0(ik^3,T)\,dk,\ \ H_1^-(x,t):=\int_{\partial D_5} e^{i\frac{k}{\sigma_L}x-ik^3t}ikM_1(ik^3,T)\,dk,\ \ (x<0), \tag{3.5a}$$

and

$$H_0^+(x,t):=\int_{\partial D_5} e^{i\frac{k}{\sigma_R}x-ik^3t}k^2M_0(ik^3,T)\,dk,\ \ H_1^+(x,t):=\int_{\partial D_5} e^{i\frac{k}{\sigma_R}x-ik^3t}ikM_1(ik^3,T)\,dk,\ \ (x>0), \tag{3.5b}$$

respectively. After splitting $S_L[u_0,\mu_0,\mu_1;f]$ and $S_R[u_0,\mu_0,\mu_1;f]$, we proceed to rewrite the functions $v_j^{\pm}(x,t)$, $w_j^{\pm}(x,t)$, $H_0^{\pm}(x,t)$, and $H_1^{\pm}(x,t)$ to facilitate the subsequent estimates. By the change of variables $k\mapsto -\alpha k$ and $k\mapsto \tilde{\alpha}k$ with $\alpha=e^{i\frac{2\pi}{3}}$ and $\tilde{\alpha}=e^{-i\frac{2\pi}{3}}$, the functions $v_j^{\pm}(x,t)$ can be further split into two parts:

$$v_j^-(x,t)=\int_0^\infty e^{-i\frac{\alpha}{\sigma_L}kx+ik^3t}\mathfrak{v}_{j,1}(k)\,dk+\int_0^\infty e^{i\frac{\tilde{\alpha}}{\sigma_L}kx-ik^3t}\mathfrak{v}_{j,2}(k)\,dk:=v_{j,1}^-(x,t)+v_{j,2}^-(x,t),\quad j=1,2,3,4, \tag{3.6a}$$

$$v_j^+(x,t)=\int_0^\infty e^{-i\frac{\alpha}{\sigma_R}kx+ik^3t}\mathfrak{v}_{j,1}(k)\,dk+\int_0^\infty e^{i\frac{\tilde{\alpha}}{\sigma_R}kx-ik^3t}\mathfrak{v}_{j,2}(k)\,dk:=v_{j,1}^+(x,t)+v_{j,2}^+(x,t),\quad j=5,6,7,8, \tag{3.6b}$$

where

$$\mathfrak{v}_{1,1}(k)=\mathfrak{v}_{5,1}(k)=\alpha\widehat{u_0^L}\left(\frac{-\alpha^2k}{\sigma_L}\right),\quad \mathfrak{v}_{1,2}(k)=\mathfrak{v}_{5,2}(k)=\tilde{\alpha}\widehat{u_0^L}\left(\frac{k}{\sigma_L}\right), \tag{3.6c}$$

$$\mathfrak{v}_{2,1}(k)=\mathfrak{v}_{6,1}(k)=\alpha\widehat{u_0^L}\left(\frac{-k}{\sigma_L}\right),\quad \mathfrak{v}_{2,2}(k)=\mathfrak{v}_{6,2}(k)=\tilde{\alpha}\widehat{u_0^L}\left(\frac{\alpha k}{\sigma_L}\right), \tag{3.6d}$$

$$\mathfrak{v}_{3,1}(k)=\mathfrak{v}_{7,1}(k)=\alpha\widehat{u_0^R}\left(\frac{-\alpha^2k}{\sigma_R}\right),\quad \mathfrak{v}_{3,2}(k)=\mathfrak{v}_{7,2}(k)=\tilde{\alpha}\widehat{u_0^R}\left(\frac{k}{\sigma_R}\right), \tag{3.6e}$$

$$\mathfrak{v}_{4,1}(k)=\mathfrak{v}_{8,1}(k)=\alpha\widehat{u_0^R}\left(\frac{-k}{\sigma_R}\right),\quad \mathfrak{v}_{4,2}(k)=\mathfrak{v}_{8,2}(k)=\tilde{\alpha}\widehat{u_0^R}\left(\frac{\alpha k}{\sigma_R}\right). \tag{3.6f}$$

Next, by using the same change of variables to the functions $w_j^{\pm}(x,t)$, we have

$$\begin{aligned}w_j^-(x,t)&=\int_0^t\int_0^\infty e^{-i\frac{\alpha}{\sigma_L}kx+ik^3(t-t')}\mathfrak{F}_{j,1}(k,t')\,dkdt'+\int_0^t\int_0^\infty e^{i\frac{\tilde{\alpha}}{\sigma_L}kx-ik^3(t-t')}\mathfrak{F}_{j,2}(k,t')\,dkdt'\\&:=w_{j,1}^-(x,t)+w_{j,2}^-(x,t),\qquad j=1,2,3,4,\end{aligned} \tag{3.7a}$$

$$\begin{aligned}w_j^+(x,t)&=\int_0^t\int_0^\infty e^{-i\frac{\alpha}{\sigma_R}kx+ik^3(t-t')}\mathfrak{F}_{j,1}(k,t')\,dkdt'+\int_0^t\int_0^\infty e^{i\frac{\tilde{\alpha}}{\sigma_R}kx-ik^3(t-t')}\mathfrak{F}_{j,2}(k,t')\,dkdt'\\&:=w_{j,1}^+(x,t)+w_{j,2}^+(x,t),\qquad j=5,6,7,8,\end{aligned} \tag{3.7b}$$

where

$$\mathfrak{F}_{1,1}(k,t')=\mathfrak{F}_{5,1}(k,t')=\alpha\widehat{f_L}\left(\frac{-\alpha^2k}{\sigma_L},t'\right),\quad \mathfrak{F}_{1,2}(k,t')=\mathfrak{F}_{5,2}(k,t')=\tilde{\alpha}\widehat{f_L}\left(\frac{k}{\sigma_L},t'\right), \tag{3.7c}$$

$$\mathfrak{F}_{2,1}(k,t')=\mathfrak{F}_{6,1}(k,t')=\alpha\widehat{f_L}\left(\frac{-k}{\sigma_L},t'\right),\quad \mathfrak{F}_{2,2}(k,t')=\mathfrak{F}_{6,2}(k,t')=\tilde{\alpha}\widehat{f_L}\left(\frac{\alpha k}{\sigma_L},t'\right), \tag{3.7d}$$

$$\mathfrak{F}_{3,1}(k,t') = \mathfrak{F}_{7,1}(k,t') = \alpha \widehat{f_R}\left(\frac{-\alpha^2 k}{\sigma_R}, t'\right), \quad \mathfrak{F}_{3,2}(k,t') = \mathfrak{F}_{7,2}(k,t') = \tilde{\alpha} \widehat{f_R}\left(\frac{k}{\sigma_R}, t'\right), \tag{3.7e}$$

$$\mathfrak{F}_{4,1}(k,t') = \mathfrak{F}_{8,1}(k,t') = \alpha \widehat{f_R}\left(\frac{-k}{\sigma_R}, t'\right), \quad \mathfrak{F}_{4,2}(k,t') = \mathfrak{F}_{8,2}(k,t') = \tilde{\alpha} \widehat{f_R}\left(\frac{\alpha k}{\sigma_R}, t'\right). \tag{3.7f}$$

Finally, by applying the same change of variables to the functions $H_0^{\pm}(x,t)$ and $H_1^{\pm}(x,t)$, we find

$$H_1^{\pm}(x,t) = H_{1,1}^{\pm}(x,t) + H_{1,2}^{\pm}(x,t), \qquad H_0^{\pm}(x,t) = H_{0,1}^{\pm}(x,t) + H_{0,2}^{\pm}(x,t), \tag{3.8a}$$

where

$$H_{1,1}^{\pm}(x,t) = -\alpha^2 \int_0^{\infty} e^{-i\frac{\alpha}{\nu^{\pm}}kx + ik^3 t} ikM_1(-ik^3, T)\,dk, \qquad H_{1,2}^{\pm}(x,t) = \tilde{\alpha}^2 \int_0^{\infty} e^{i\frac{\tilde{\alpha}}{\nu^{\pm}}kx - ik^3 t} ikM_1(ik^3, T)\,dk, \tag{3.8b}$$

$$H_{0,1}^{\pm}(x,t) = \int_0^{\infty} e^{-i\frac{\alpha}{\nu^{\pm}}kx + ik^3 t} k^2 M_0(-ik^3, T)\,dk, \qquad H_{0,2}^{\pm}(x,t) = \int_0^{\infty} e^{i\frac{\tilde{\alpha}}{\nu^{\pm}}kx - ik^3 t} k^2 M_0(ik^3, T)\,dk. \tag{3.8c}$$

Here, the parameter $\nu^{\pm}$ is determined by the region of x. Specifically, $\nu^- = \sigma_L$ when $x < 0$, and $\nu^+ = \sigma_R$ when $x > 0$. Observe that all the functions $v_j^{\pm}(x,t)$ include similar exponential terms, with differences arising only from the coefficients σ_L, σ_R, and the functions $\mathfrak{v}_{j,1}(k)$, $\mathfrak{v}_{j,2}(k)$. This observation implies that we only need to consider the estimates of $v_{1,1}^-(x,t)$ and $v_{1,2}^-(x,t)$. The estimates for the other terms $v_{j,1}^{\pm}(x,t)$ and $v_{j,2}^{\pm}(x,t)$ can be derived using the same method. Furthermore, when analyzing the functions $w_j^{\pm}(x,t)$, we apply Minkowski's integral inequality and follow a similar approach as in the estimates for $v_j^{\pm}(x,t)$ to achieve the desired results. Finally, for the estimates of $H_0^{\pm}(x,t)$ and $H_1^{\pm}(x,t)$, we can derive similar upper bounds by adopting the method used for $v_1^{\pm}(x,t)$ while replacing the functions $\mathfrak{v}_{1,1}(k)$ and $\mathfrak{v}_{1,2}(k)$ with $k^2 M_0$ and ikM_1, respectively.

3.1 | The Estimates of the Functions (3.2a) and (3.2b)

In this subsection, we establish the following estimates for the functions $\mathcal{U}_L(x,t)$ and $\mathcal{F}_L(x,t)$:

$$\sup_{t\in[0,T]} \|\mathcal{U}_L(\cdot,t)\|_{H^s(\mathbb{R}^-)} + \sup_{t\in[0,T]} \|\mathcal{F}_L(\cdot,t)\|_{H^s(\mathbb{R}^-)} \lesssim \|u_0^L\|_{H^s(\mathbb{R}^-)} + \int_0^T \|f_L(\cdot,t)\|_{H^s(\mathbb{R}^-)} dt, \qquad s \geq 0. \tag{3.9a}$$

Moreover, we have

$$\sup_{x\in\mathbb{R}^-} \|\mathcal{U}_L(x,\cdot)\|_{H^{(s+1)/3}((0,T))} \lesssim_{s,\sigma_L} \|u_0^L\|_{H^s(\mathbb{R}^-)}, \qquad -1 \leq s < 2, \tag{3.9b}$$

$$\sup_{x\in\mathbb{R}^-} \|(\mathcal{U}_L)_x(x,\cdot)\|_{H^{s/3}((0,T))} \lesssim_{s,\sigma_L} \|u_0^L\|_{H^s(\mathbb{R}^-)}, \qquad 0 \leq s < 3, \tag{3.9c}$$

and

$$\sup_{x\in\mathbb{R}^-} \|\mathcal{F}_L(x,\cdot)\|_{H^{(s+1)/3}((0,T))} \lesssim_{s,\sigma_L} \begin{cases} \max\left\{T^{\frac{1}{2}}, T^{\frac{1-2s}{6}}\right\} \|f_L\|_{L^2([0,T];H^s(\mathbb{R}^-))}, & -1 \leq s < \frac{1}{2}, \\ \max\left\{T^{\frac{1}{2}}, T^{\frac{2-s}{3}}\right\} \|f_L\|_{L^2([0,T];H^s(\mathbb{R}^-))}, & \frac{1}{2} < s < 2, \end{cases} \tag{3.9d}$$

$$\sup_{x\in\mathbb{R}^-} \|(\mathcal{F}_L)_x(x,\cdot)\|_{H^{s/3}((0,T))} \lesssim_{s,\sigma_L} \begin{cases} \max\left\{T^{\frac{1}{2}}, T^{\frac{3-2s}{6}}\right\} \|f_L\|_{L^2([0,T];H^s(\mathbb{R}^-))}, & 0 \leq s < \frac{3}{2}, \\ \max\left\{T^{\frac{1}{2}}, T^{\frac{3-s}{3}}\right\} \|f_L\|_{L^2([0,T];H^s(\mathbb{R}^-))}, & \frac{3}{2} < s < 3. \end{cases} \tag{3.9e}$$

As a consequence of (3.9), the estimates for $\mathcal{U}_R(x,t)$ and $\mathcal{F}_R(x,t)$ can be established in an analogous manner.

Proof of (3.9a). Let $U_0^L(x)$ be an extension function of $u_0^L(x)$ satisfying

$$U_0^L(x)\Big|_{\mathbb{R}^-} = u_0^L(x), \qquad \|U_0^L\|_{H^s(\mathbb{R})} \lesssim \|u_0^L\|_{H^s(\mathbb{R}^-)}. \tag{3.10}$$

Thus, we have

$$\|\mathcal{U}_L(\cdot,t)\|^2_{H^s(\mathbb{R}^-)} \leq \int_{\mathbb{R}} (1+k^2)^s |\widehat{U_0^L}(k)|^2 dk \lesssim \|u_0^L\|^2_{H^s(\mathbb{R}^-)}. \tag{3.11}$$

In addition, we let $f_e(x,t)$ be the extension function of $f_L(x,t)$ satisfying

$$f_e(x,t)|_{\mathbb{R}^-} = f_L(x,t), \qquad \|f_e(\cdot,t)\|_{H^s(\mathbb{R})} \lesssim \|f_L(\cdot,t)\|_{H^s(\mathbb{R}^-)}, \qquad \text{for } t>0. \tag{3.12}$$

Applying Minkowski's integral inequality and the same estimate in (3.11), we obtain

$$\|\mathcal{F}_L(\cdot,t)\|_{H^s(\mathbb{R}^-)} \lesssim \int_0^T \|f_e(\cdot,t)\|_{H^s(\mathbb{R})} dt \lesssim \int_0^T \|f_L(\cdot,t)\|_{H^s(\mathbb{R}^-)} dt. \tag{3.13}$$

Combining the estimates (3.11) and (3.13), we conclude that (3.9a) holds. □

Proof of (3.9b). For convenience, we denote $m_0 = (s+1)/3$, and we shall use this notation throughout the remainder of the paper. To establish the estimate (3.9b), we begin by introducing a smooth cut-off function $\theta_1(k)$ and define the function $\tilde{U}_L(x,t)$ as follows:

$$\theta_1(k) = \begin{cases} 1, & |k| \leq \frac{1}{\sigma_L}, \\ 0, & |k| \geq \frac{2}{\sigma_L}, \end{cases} \tag{3.14}$$

and

$$\tilde{U}_L(x,t) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{ikx - ik^3\sigma_L^3 t}\widehat{U_0^L}(k)\, dk = \mathrm{I}_1^{(U)}(x,t) + \mathrm{I}_2^{(U)}(x,t), \tag{3.15}$$

where

$$\mathrm{I}_1^{(U)}(x,t) := \frac{1}{2\pi}\int_{\mathbb{R}} e^{ikx-ik^3\sigma_L^3 t}\theta_1(k)\widehat{U_0^L}(k)\, dk, \quad \mathrm{I}_2^{(U)}(x,t) := \frac{1}{2\pi}\int_{\mathbb{R}} e^{ikx-ik^3\sigma_L^3 t}(1-\theta_1(k))\widehat{U_0^L}(k)\, dk. \tag{3.16}$$

For any $j = 0,1,2,\ldots$, we have

$$\partial_t^j \mathrm{I}_1^{(U)}(x,t) := \frac{1}{2\pi}\int_{\mathbb{R}} e^{ikx-ik^3\sigma_L^3 t}\theta_1(k)(-ik^3\sigma_L^3)^j\,\widehat{U_0^L}(k)\, dk. \tag{3.17}$$

This expression can be rewritten as

$$\partial_t^j \mathrm{I}_1^{(U)}(x,t) = \frac{1}{2\pi}\int_{-\frac{2}{\sigma_L}}^{\frac{2}{\sigma_L}} \left[e^{ikx-ik^3\sigma_L^3 t}\theta_1(k)(-ik^3\sigma_L^3)^j (1+k^2)^{\frac{-s}{2}}\right]\left[(1+k^2)^{\frac{s}{2}}\widehat{U_0^L}(k)\right] dk. \tag{3.18}$$

Applying Hölder inequality, we obtain

$$|\partial_t^j \mathrm{I}_1^{(U)}(x,t)| \leq C\left(\int_{-\frac{2}{\sigma_L}}^{\frac{2}{\sigma_L}} \frac{|\theta_1(k)|^2|k|^{6j}}{(1+k^2)^s}\right)^{\frac{1}{2}} \|U_0^L\|_{H^s(\mathbb{R})} \lesssim_{s,\sigma_L,j} \|u_0^L\|_{H^s(\mathbb{R}^-)}. \tag{3.19}$$

Consequently,

$$\|\partial_t^j \mathrm{I}_1^{(U)}(x,\cdot)\|_{L^2((0,T))} \lesssim_{s,\sigma_L,j} T^{\frac{1}{2}}\|u_0^L\|_{H^s(\mathbb{R}^-)}. \tag{3.20}$$

Next, we address the case when the index m_0 is noninteger. To this end, we recall the following Sobolev interpolation inequality.

Lemma 3.1. *Let $m_1 < m < m_2$. Then*

$$\|f\|_{H^m} \leq \|f\|_{H^{m_1}}^{\frac{m_2-m}{m_2-m_1}} \|f\|_{H^{m_2}}^{\frac{m-m_1}{m_2-m_1}}. \tag{3.21}$$

Applying Lemma 3.1 with $0 < m_0 < 1$, we find that

$$\|\mathrm{I}_1^{(U)}(x,\cdot)\|_{H^{m_0}((0,T))} \leq \|\mathrm{I}_1^{(U)}(x,\cdot)\|_{L^2((0,T))}^{1-m_0} \|\mathrm{I}_1^{(U)}(x,\cdot)\|_{H^1((0,T))}^{m_0} \lesssim_{s,\sigma_L} T^{\frac{1}{2}} \|u_0^L\|_{H^s(\mathbb{R}^-)}. \tag{3.22}$$

We now examine the second term, $\mathrm{I}_2^{(U)}(x,t)$, and rewrite it as

$$\mathrm{I}_2^{(U)}(x,t) = \frac{1}{2\pi}\int_{-\infty}^{-\frac{1}{\sigma_L}} e^{ikx-ik^3\sigma_L^3 t}(1-\theta_1(k))\widehat{U_0^L}(k)\,dk + \frac{1}{2\pi}\int_{\frac{1}{\sigma_L}}^{\infty} e^{ikx-ik^3\sigma_L^3 t}(1-\theta_1(k))\widehat{U_0^L}(k)\,dk := \mathrm{I}_{2,1}^{(U)}(x,t) + \mathrm{I}_{2,2}^{(U)}(x,t). \tag{3.23}$$

To analyze the integral $\mathrm{I}_{2,1}^{(U)}(x,t)$, we apply the change of variable $k = -\tau^{1/3}/\sigma_L$ to obtain

$$\mathrm{I}_{2,1}^{(U)}(x,t) = \frac{1}{2\pi}\int_1^{\infty} e^{i\tau t}\left[e^{-i\frac{\tau^{\frac{1}{3}}}{\sigma_L}x}\left(1-\theta_1\left(\frac{-\tau^{\frac{1}{3}}}{\sigma_L}\right)\right)\widehat{U_0^L}\left(\frac{-\tau^{\frac{1}{3}}}{\sigma_L}\right)\frac{\tau^{\frac{-2}{3}}}{3\sigma_L}\right]d\tau. \tag{3.24}$$

The associated Fourier transform of $\mathrm{I}_{2,1}^{(U)}(x,t)$ in time is given by

$$\widehat{\mathrm{I}_{2,1}^{(U)}}(x,\tau) = \begin{cases} e^{-i\frac{\tau^{\frac{1}{3}}}{\sigma_L}x}\left(1-\theta_1\left(\frac{-\tau^{\frac{1}{3}}}{\sigma_L}\right)\right)\widehat{U_0^L}\left(\frac{-\tau^{\frac{1}{3}}}{\sigma_L}\right)\frac{\tau^{\frac{-2}{3}}}{3\sigma_L}, & \tau \in (1,\infty), \\ 0, & \tau \in (-\infty, 1). \end{cases} \tag{3.25}$$

Consequently,

$$\|\mathrm{I}_{2,1}^{(U)}(x,\cdot)\|_{H^{m_0}((0,T))}^2 \leq \int_{\mathbb{R}} (1+\tau^2)^{m_0} |\widehat{\mathrm{I}_{2,1}^{(U)}}(x,\tau)|^2\,d\tau \lesssim_s \int_1^{\infty} |\tau|^{2m_0-\frac{4}{3}}\left|\widehat{U_0^L}\left(\frac{-\tau^{\frac{1}{3}}}{\sigma_L}\right)\right|^2 d\tau \lesssim_{s,\sigma_L} \|U_0^L\|_{H^s(\mathbb{R})}^2 \lesssim \|u_0^L\|_{H^s(\mathbb{R}^-)}^2. \tag{3.26}$$

A similar argument applies to $\mathrm{I}_{2,2}^{(U)}(x,t)$. By the change of variable $k = (-\tau)^{1/3}/\sigma_L$, we obtain

$$\|\mathrm{I}_{2,2}^{(U)}(x,\cdot)\|_{H^{m_0}((0,T))}^2 \lesssim_s \int_{-\infty}^{-1} |\tau|^{2m_0-\frac{4}{3}}\left|\widehat{U_0^L}\left(\frac{(-\tau)^{\frac{1}{3}}}{\sigma_L}\right)\right|^2 d\tau \lesssim_{s,\sigma_L} \|u_0^L\|_{H^s(\mathbb{R}^-)}^2. \tag{3.27}$$

Combining the estimates (3.22), (3.26), and (3.27), we conclude that for $0 < m_0 < 1$,

$$\sup_{x\in\mathbb{R}^-} \|\mathcal{U}_L(x,\cdot)\|_{H^{m_0}((0,T))} \leq \sup_{x\in\mathbb{R}^-} \|\mathrm{I}_1^{(U)}(x,\cdot)\|_{H^{m_0}((0,T))} + \sum_{j=1}^{2}\sup_{x\in\mathbb{R}^-} \|\mathrm{I}_{2,j}^{(U)}(x,\cdot)\|_{H^{m_0}((0,T))} \lesssim_{s,\sigma_L} \|u_0^L\|_{H^s(\mathbb{R}^-)}. \tag{3.28}$$

□

Proof of (3.9c). The proof of (3.9c) is similar to that of (3.9b). The only difference is that, when considering the function $(\mathcal{U}_L)_x$, the Sobolev exponent used becomes $s/3$. With this modification, the estimate (3.9c) follows from the same argument as in the proof of (3.9b). □

Proof of (3.9d). Note that for $0 < m_0 < 1$,

$$\|\mathcal{F}_L(x,\cdot)\|^2_{H^{m_0}((0,T))} = \|\mathcal{F}_L(x,\cdot)\|^2_{L^2((0,T))} + [\mathcal{F}_L(x,\cdot)]^2_{m_0}. \tag{3.29}$$

Applying Minkowski's integral inequality and the same estimate in (3.9b), the L^2 norm of the function $\mathcal{F}_L(x,t)$ can be bounded as follows:

$$\|\mathcal{F}_L(x,\cdot)\|^2_{L^2((0,T))} \lesssim_{s,\sigma_L} \left(\int_0^T \|f_e(\cdot,t)\|_{H^s(\mathbb{R})}\right)^2 \lesssim T\|f_L\|^2_{L^2([0,T];H^s(\mathbb{R}^-))}. \tag{3.30}$$

Note also that

$$[\mathcal{F}_L(x,\cdot)]^2_{m_0} := \int_0^T\int_0^T \frac{|\mathcal{F}_L(x,t)-\mathcal{F}_L(x,t')|^2}{|t-t'|^{1+2m_0}}dtdt' = 2\int_0^T\int_0^{T-t}\frac{|\mathcal{F}_L(x,t+z)-\mathcal{F}_L(x,t)|^2}{z^{1+2m_0}}dzdt \tag{3.31}$$

and

$$\begin{aligned}|\mathcal{F}_L(x,t+z)-\mathcal{F}_L(x,t)|^2 \lesssim{}& \left|\frac{1}{2\pi}\int_0^t\int_{\mathbb{R}} e^{ikx-ik^3\sigma_L^3(t-t')}(e^{-ik^3\sigma_L^3 z}-1)\widehat{f_L}(k,t')\,dkdt'\right|^2\\ &+\left|\frac{1}{2\pi}\int_t^{t+z}\int_{\mathbb{R}} e^{ikx-ik^3\sigma_L^3(t+z-t')}\widehat{f_L}(k,t')dkdt'\right|^2.\end{aligned} \tag{3.32}$$

We estimate these two terms separately. Applying Minkowski's inequality to switch $dzdt$ and dt' for the first term of (3.32), we find

$$\begin{aligned}&\int_0^T\int_0^{T-t}\frac{1}{z^{1+2m_0}}\left|\frac{1}{2\pi}\int_0^t\int_{\mathbb{R}} e^{ikx-ik^3\sigma_L^3(t-t')}(e^{-ik^3\sigma_L^3 z}-1)\widehat{f_L}(k,t')\,dkdt'\right|^2 dzdt\\ &\lesssim\left(\int_0^T\left(\int_0^T\int_0^{T-t}\frac{1}{z^{1+2m_0}}\left|\int_{\mathbb{R}} e^{ikx-ik^3\sigma_L^3(t-t')}(e^{-ik^3\sigma_L^3 z}-1)\widehat{f_L}(k,t')\,dk\right|^2 dzdt\right)^{\frac12}dt'\right)^2.\end{aligned} \tag{3.33}$$

Thus, using the same estimate as in (3.9b) and the inequality (3.12), we obtain

$$\int_0^T\int_0^{T-t}\frac{1}{z^{1+2m_0}}\left|\frac{1}{2\pi}\int_0^t\int_{\mathbb{R}} e^{ikx-ik^3\sigma_L^3(t-t')}(e^{-ik^3\sigma_L^3 z}-1)\widehat{f_L}(k,t')\,dkdt'\right|^2 dkdt' \lesssim_{s,\sigma_L} T\|f_L\|^2_{L^2([0,T];H^s(\mathbb{R}^-))}. \tag{3.34}$$

For the second term of (3.32), we consider two different cases: $0 < m_0 < 1/2$ and $1/2 < m_0 < 1$.

Case 1. $0 < m_0 < 1/2$. Using Hölder inequality, we find

$$\begin{aligned}&\int_0^T\int_0^{T-t}\frac{1}{z^{1+2m_0}}\left|\frac{1}{2\pi}\int_t^{t+z}\int_{\mathbb{R}} e^{ikx-ik^3\sigma_L^3(t+z-t')}\widehat{f_L}(k,t')dkdt'\right|^2 dzdt\\ &\lesssim\int_0^T\int_0^{T-t}\frac{1}{z^{2m_0}}\left(\int_t^{t+z}\left|\int_{\mathbb{R}} e^{ikx-ik^3\sigma_L^3(t+z-t')}\widehat{f_L}(k,t')dk\right|^2 dt'\right)dzdt.\end{aligned} \tag{3.35}$$

Applying Fubini's theorem twice, we find

$$\int_0^T\int_0^{T-t}\frac{1}{z^{2m_0}}\left(\int_t^{t+z}\left|\int_{\mathbb{R}}e^{ikx-ik^3\sigma_L^3(t+z-t')}\widehat{f_L}(k,t')dk\right|^2dt'\right)dzdt$$

$$=\int_0^T\frac{1}{z^{2m_0}}\left(\int_0^T\int_{t'-z}^{T-z}\left|\int_{\mathbb{R}}e^{ikx-ik^3\sigma_L^3(t+z-t')}\widehat{f_L}(k,t')dk\right|^2dtdt'\right)dz. \tag{3.36}$$

By the change of variable $t_1=t+z$, we get

$$\int_0^T\frac{1}{z^{2m_0}}\left(\int_0^T\left\|\int_{\mathbb{R}}e^{ikx-ik^3\sigma_L^3(t_1-t')}\widehat{f_L}(k,t')dk\right\|^2_{L^2((t',T))}dt'\right)dz. \tag{3.37}$$

Therefore, we show that for $0<m_0<1/2$,

$$\int_0^T\int_0^{T-t}\frac{1}{z^{1+2m_0}}\left|\frac{1}{2\pi}\int_t^{t+z}\int_{\mathbb{R}}e^{ikx-ik^3\sigma_L^3(t+z-t')}\widehat{f_L}(k,t')dkdt'\right|^2dzdt$$

$$\lesssim_s T^{\frac{1-2s}{3}}\int_0^T\left\|\int_{\mathbb{R}}e^{ikx-ik^3\sigma_L^3(t_1-t')}\widehat{f_L}(k,t')dk\right\|^2_{L^2((0,T))}dt'$$

$$\lesssim_{s,\sigma_L}T^{\frac{1-2s}{3}}\int_0^T\|f_L(\cdot,t')\|^2_{H^s(\mathbb{R}^-)}dt', \tag{3.38}$$

where we have used the estimate (3.9b) and the estimate (3.12) in the last inequality.

Case 2. $1/2<m_0<1$. Observe that

$$\int_0^T\int_0^{T-t}\frac{1}{z^{1+2m_0}}\left|\frac{1}{2\pi}\int_t^{t+z}\int_{\mathbb{R}}e^{ikx-ik^3\sigma_L^3(t+z-t')}\widehat{f_L}(k,t')\,dkdt'\right|^2dzdt$$

$$=\int_0^T\int_0^{T-t}\frac{1}{z^{1+2m_0}}\left|\frac{1}{2\pi}\int_{\mathbb{R}}e^{ikx-ik^3\sigma_L^3(t+z)}\int_t^{t+z}e^{ik^3\sigma_L^3t'}\widehat{f_L}(k,t')\,dt'dk\right|^2dzdt. \tag{3.39}$$

Since $m_0>1/2$, we apply the Sobolev inequality

$$\|\varphi\|_{L^\infty}\lesssim_s\|\varphi\|_{H^s},\qquad s>\frac{1}{2} \tag{3.40}$$

to obtain

$$\int_0^T\int_0^{T-t}\frac{1}{z^{1+2m_0}}\left|\frac{1}{2\pi}\int_{\mathbb{R}}e^{ikx-ik^3\sigma_L^3(t+z)}\int_t^{t+z}e^{ik^3\sigma_L^3t'}\widehat{f_L}(k,t')\,dt'dk\right|^2dzdt$$

$$\lesssim_s\int_0^T\int_0^{T-t}\frac{1}{z^{1+2m_0}}\left[\int_{\mathbb{R}}(1+k^2)^s\left|\int_t^{t+z}e^{ik^3\sigma_L^3t'}\widehat{f_L}(k,t')dt'\right|^2dk\right]dzdt. \tag{3.41}$$

Using Minkowski's integral inequality to switch dk and dt', we get

$$\int_0^T\int_0^{T-t}\frac{1}{z^{1+2m_0}}\left[\int_{\mathbb{R}}(1+k^2)^s\left|\int_t^{t+z}e^{ik^3\sigma_L^3t'}\widehat{f_L}(k,t')dt'\right|^2dk\right]dzdt\le\int_0^T\int_0^{T-t}\frac{1}{z^{1+2m_0}}\left(\int_t^{t+z}\|f_e(\cdot,t')\|_{H^s(\mathbb{R})}\right)^2dzdt. \tag{3.42}$$

Finally, by applying Hölder inequality and Fubini's theorem twice, we find

$$\begin{aligned}\int_0^T\int_0^{T-t}\frac{1}{z^{1+2m_0}}\left(\int_t^{t+z}\|f_e(\cdot,t')\|_{H^s(\mathbb{R})}\right)^2dzdt &\lesssim \int_0^T\int_0^{T-t}\int_t^{t+z} z^{-2m_0}\|f_e(\cdot,t')\|^2_{H^s(\mathbb{R})}dt'dzdt\\ &= \int_0^T\|f_e(\cdot,t')\|^2_{H^s(\mathbb{R})}\left(\int_0^T z^{-2m_0}\int_{t'-z}^{t'}dtdz\right)dt'\\ &\lesssim_s T^{\frac{4-2s}{3}}\int_0^T\|f_L(\cdot,t)\|^2_{H^s(\mathbb{R}^-)}dt'. \end{aligned}\tag{3.43}$$

Combining the estimates (3.30), (3.31), (3.34), (3.38), and (3.43), we obtain the desired result (3.9d). □

Proof of the inequality (3.9e). The proof of (3.9e) is similar to that of (3.9d). The only difference is that, when considering the function $(\mathcal{F}_L)_x$, the Sobolev exponent used becomes $s/3$. With this modification, the estimate (3.9e) follows the same argument as in the proof of (3.9d). □

3.2 | The Estimates of the Functions (3.6a) and (3.6b)

In this subsection, we establish the following estimates for the functions $v_j^{\pm}(x,t)$:

$$\sup_{t\in[0,T]}\|v_j^{\pm}(\cdot,t)\|_{H^s(\mathbb{R}^{\pm})}\lesssim_{s,\sigma_L,\sigma_R}\|u_0^L\|_{H^s(\mathbb{R}^-)}+\|u_0^R\|_{H^s(\mathbb{R}^+)},\qquad s\geq 0.\tag{3.44a}$$

Moreover, we can show that

$$\sup_{x\in\mathbb{R}^{\pm}}\|v_j^{\pm}(x,\cdot)\|_{H^{(s+1)/3}((0,T))}\lesssim_{s,\sigma_L,\sigma_R}(1+T^{\frac{1}{2}})\big(\|u_0^L\|_{H^s(\mathbb{R}^-)}+\|u_0^R\|_{H^s(\mathbb{R}^+)}\big),\qquad -1\leq s<2\tag{3.45a}$$

$$\sup_{x\in\mathbb{R}^{\pm}}\|(v_j^{\pm})_x(x,\cdot)\|_{H^{s/3}((0,T))}\lesssim_{s,\sigma_L,\sigma_R}(1+T^{\frac{1}{2}})\big(\|u_0^L\|_{H^s(\mathbb{R}^-)}+\|u_0^R\|_{H^s(\mathbb{R}^+)}\big),\qquad 0\leq s<3.\tag{3.45b}$$

Proof of (3.44a). According to the definitions of (3.6), it suffices to prove the estimate for $v_1^-(x,t)$. The proofs for the other functions $v_j^{\pm}(x,t)$ are analogous.

We split the proof into three cases: $\lfloor s\rfloor=0$; $\beta=0$; $\lfloor s\rfloor\neq 0$ and $\beta\neq 0$.

Case 1. $\lfloor s\rfloor=0$. We have

$$\left[v_{1,1}^-(\cdot,t)\right]^2_{\beta-}=\int_{-\infty}^0\int_{-\infty}^0\frac{|v_{1,1}^-(x,t)-v_{1,1}^-(y,t)|^2}{|x-y|^{1+2\beta}}dxdy=\int_0^\infty\int_0^\infty\frac{|v_{1,1}^-(-x,t)-v_{1,1}^-(-y,t)|^2}{|x-y|^{1+2\beta}}dxdy,\tag{3.46}$$

where

$$v_{1,1}^-(-x,t)=\int_0^\infty e^{i\frac{\alpha}{\sigma_L}kx}\left[e^{ik^3t}\mathfrak{v}_{1,1}(k)\right]dk.\tag{3.47}$$

The following lemma provides a way of handling the complex exponentials in $|v_{1,1}^-(-x,t)-v_{1,1}^-(-y,t)|$. The proof of Lemma 3.2 below can be found in [22].

Lemma 3.2. *Let $k,x,y\geq 0$. For any complex number γ, we have the following inequality*

$$\left|e^{i\gamma kx}-e^{i\gamma ky}\right|\leq\sqrt{2}\left(1+\frac{|\gamma_R|}{|\gamma_I|}\right)\left|e^{-\gamma_I kx}-e^{-\gamma_I ky}\right|,\tag{3.48}$$

where $\gamma=\gamma_R+i\gamma_I$.

By applying Lemma 3.2 with $\gamma = \alpha/\sigma_L$ and defining $G_{1,1}(k,t) := e^{ik^3 t}\mathfrak{v}_{1,1}(k)$, we find

$$\begin{aligned}\left|v_{1,1}^{-}(-x,t) - v_{1,1}^{-}(-y,t)\right|^2 &\le \int_0^\infty \int_0^\infty \left|e^{i\gamma kx} - e^{i\gamma ky}\right|\left|e^{i\gamma \xi x} - e^{i\gamma \xi y}\right| |G_{1,1}(\xi,t)||G_{1,1}(k,t)|\, dk\, d\xi \\ &\lesssim_{\sigma_L} \int_0^\infty \int_0^\infty \left|e^{-\frac{\sqrt{3}}{2\sigma_L}kx} - e^{-\frac{\sqrt{3}}{2\sigma_L}ky}\right|\left|e^{-\frac{\sqrt{3}}{2\sigma_L}\xi x} - e^{-\frac{\sqrt{3}}{2\sigma_L}\xi y}\right| |G_{1,1}(\xi,t)||G_{1,1}(k,t)|\, dk\, d\xi. \end{aligned} \tag{3.49}$$

Therefore,

$$\left[v_{1,1}^{-}(\cdot,t)\right]_{\beta-}^2 \lesssim \int_0^\infty \int_0^\infty |G_{1,1}(k,t)||G_{1,1}(\xi,t)| \bigtriangleup \, dk d\xi, \tag{3.50}$$

where we define

$$\bigtriangleup := \int_0^\infty \int_0^\infty \frac{(e^{-\frac{\sqrt{3}}{2\sigma_L}kx} - e^{-\frac{\sqrt{3}}{2\sigma_L}ky})(e^{-\frac{\sqrt{3}}{2\sigma_L}\xi x} - e^{-\frac{\sqrt{3}}{2\sigma_L}\xi y})}{|x-y|^{1+2\beta}}\, dx dy. \tag{3.51}$$

Let $a = \left(k\xi^{-1}\right)^{1/2}$. By the change of variables

$$x' = \frac{\sqrt{3}}{2\sigma_L}(k\xi)^{\frac{1}{2}}x, \quad y' = \frac{\sqrt{3}}{2\sigma_L}(k\xi)^{\frac{1}{2}}y, \tag{3.52}$$

the integral $\bigtriangleup$ can be written as

$$\begin{aligned}\bigtriangleup &= \left(\frac{\sqrt{3}}{2\sigma_L}\right)^{2\beta-1} (k\xi)^{\beta-\frac{1}{2}} \int_0^\infty \int_0^\infty \frac{(e^{-ax'} - e^{-ay'})(e^{-\frac{x'}{a}} - e^{-\frac{y'}{a}})}{|x'-y'|^{1+2\beta}} dx'dy' \\ &= 2\left(\frac{\sqrt{3}}{2\sigma_L}\right)^{2\beta-1} (k\xi)^{\beta-\frac{1}{2}} \int_0^\infty \int_0^\infty \frac{e^{-\left(a+\frac{1}{a}\right)x} - e^{-\left(ay+\frac{x}{a}\right)}}{|x-y|^{1+2\beta}} dx dy. \end{aligned} \tag{3.53}$$

The following lemma provides an important estimate of the integral (3.53), and full details can be found in [21].

Lemma 3.3 (Lemma 3.1 in [21]). *Define*

$$\bigtriangleup\left(a, \frac{1}{a}, \beta\right) = \int_0^\infty \int_0^\infty \frac{e^{-\left(a+\frac{1}{a}\right)x} - e^{-\left(ay+\frac{x}{a}\right)}}{|x-y|^{1+2\beta}} dx dy. \tag{3.54}$$

If $a \in [0,\infty]$ and $\beta \in (0,1)$, then

$$\bigtriangleup\left(a, \frac{1}{a}, \beta\right) = C(\beta)\left(a + \frac{1}{a}\right)^{-1}\left(a^{2\beta} + \left(\frac{1}{a}\right)^{2\beta} - \left(a + \frac{1}{a}\right)^{2\beta}\right), \tag{3.55}$$

where

$$C(\beta) = \begin{cases} \frac{\Gamma(1-2\beta)}{\beta}, & \beta \in \left(0, \frac{1}{2}\right), \\ \frac{\pi}{\sin(2\pi\beta)\Gamma(2\beta)\beta}, & \beta \in \left[\frac{1}{2}, 1\right), \end{cases} \tag{3.56}$$

and it implies

$$\bigtriangleup\left(a, \frac{1}{a}, \beta\right) \lesssim_\beta \left(a + \frac{1}{a}\right)^{-1}, \qquad \beta \in (0,1). \tag{3.57}$$

Combining the estimates (3.50) and (3.57), we find

$$\begin{aligned}\left[v_{1,1}^{-}(\cdot,t)\right]_{\beta-}^{2} &\lesssim_{\beta,\sigma_L} \int_0^\infty\int_0^\infty |G_{1,1}(k,t)||G_{1,1}(\xi,t)|\frac{(k\xi)^\beta}{k+\xi}\,dkd\xi\\ &= \int_0^\infty\int_0^\infty |k\xi|^\beta |G_{1,1}(k,t)||G_{1,1}(\xi,t)|\left(\int_0^\infty e^{-(k+\xi)x}dx\right)dkd\xi\\ &\le \int_0^\infty\left(\int_0^\infty e^{-kx}\left((1+|k|^2)^{\frac{\beta}{2}}\left|G_{1,1}(k,t)\right|\right)dk\right)^2 dx. \end{aligned} \tag{3.58}$$

To estimate (3.58), we refer to the following lemma, proved in Lemma 3.2 of [21].

Lemma 3.4 (L^2 boundness of Laplace transformation). *If $f \in L^2(\mathbb{R}^+)$, then the integral*

$$\int_0^\infty e^{-kx} f(k)\,dk \tag{3.59}$$

is bounded from $L^2(\mathbb{R}^+)$ to $L^2(\mathbb{R}^+)$ with

$$\left\|\int_0^\infty e^{-kx} f(k)\,dk\right\|_{L^2_x(\mathbb{R}^+)} \lesssim \|f\|_{L^2(\mathbb{R}^+)}. \tag{3.60}$$

By using Lemma 3.4 with $f(k) = (1+k^2)^{\beta/2}\left|G_{1,1}(k,t)\right|$, we obtain

$$\left[v_{1,1}^{-}(\cdot,t)\right]_{\beta-} \lesssim_{\beta,\sigma_L} \|U_0^L\|_{H^\beta(\mathbb{R})} \lesssim_{\beta,\sigma_L} \|u_0^L\|_{H^\beta(\mathbb{R}^-)}. \tag{3.61}$$

Case 2. $\beta = 0$. Note that

$$\left\|v_{1,1}^{-}(\cdot,t)\right\|_{H^s(\mathbb{R}^-)}^2 = \sum_{j=0}^{\lfloor s\rfloor}\left\|\partial_x^j v_{1,1}^{-}(\cdot,t)\right\|_{L^2(\mathbb{R}^-)}^2, \tag{3.62}$$

where

$$\partial_x^j v_{1,1}^{-}(x,t) = \int_0^\infty e^{-i\frac{\alpha}{\sigma_L}kx+ik^3t}\left(-i\frac{\alpha}{\sigma_L}k\right)^j \mathfrak{v}_{1,1}(k)\,dk. \tag{3.63}$$

Let

$$\tilde{G}_{1,1}^{(j)}(k,t) = e^{ik^3t}\left(-i\frac{\alpha}{\sigma_L}k\right)^j \mathfrak{v}_{1,1}(k). \tag{3.64}$$

By using Lemma 3.4, we find

$$\left\|\partial_x^j v_{1,1}^{-}(\cdot,t)\right\|_{L^2(\mathbb{R}^-)}^2 \lesssim_{\sigma_L} \int_0^\infty\left(\int_0^\infty e^{-\frac{\sqrt{3}}{2\sigma_L}kx}|\tilde{G}_{1,1}^{(j)}(k,t)|\,dk\right)^2 dx \lesssim_{\sigma_L,j} \|U_0^L\|_{H^j(\mathbb{R}^-)}^2 \lesssim_{\sigma_L,j} \|u_0^L\|_{H^j(\mathbb{R}^-)}^2. \tag{3.65}$$

Case 3. $\lfloor s\rfloor \neq 0$ and $\beta \neq 0$. Recall that

$$\left\|v_{1,1}^{-}(\cdot,t)\right\|_{H^s(\mathbb{R}^-)}^2 = \sum_{j=0}^{\lfloor s\rfloor}\left\|\partial_x^j v_{1,1}^{-}(\cdot,t)\right\|_{L^2(\mathbb{R}^-)}^2 + \left[\partial_x^{\lfloor s\rfloor} v_{1,1}^{-}(\cdot,t)\right]_{\beta-}^2. \tag{3.66}$$

Since

$$\partial_x^{\lfloor s \rfloor} v_{1,1}^-(x,t) = \int_0^\infty e^{-i\frac{\alpha}{\sigma_L}kx} \tilde{G}_{1,1}^{(\lfloor s \rfloor)}(k,t)\, dk, \tag{3.67}$$

we follow the same approach in (3.49) to obtain

$$\left[\partial_x^{\lfloor s \rfloor} v_{1,1}^-(\cdot,t)\right]_{\beta-}^2 \lesssim_\beta \int_0^\infty \int_0^\infty |\tilde{G}_{1,1}^{(\lfloor s \rfloor)}(k,t)||\tilde{G}_{1,1}^{(\lfloor s \rfloor)}(\xi,t)| \triangle \, dk d\xi \lesssim_{s,\sigma_L} \|U_0^L\|_{H^s(\mathbb{R}^-)} \lesssim \|u_0^L\|_{H^s(\mathbb{R}^-)}. \tag{3.68}$$

It remains to prove the estimate for $v_{1,2}^-(x,t)$. We begin by considering the case $\lfloor s \rfloor = 0$. By using Lemma 3.2 with $\gamma = -\tilde{\alpha}/\sigma_L$, we find

$$\left| e^{-i\frac{\tilde{\alpha}}{\sigma_L}kx} - e^{-i\frac{\tilde{\alpha}}{\sigma_L}ky} \right| \lesssim_{\sigma_L} |e^{-\frac{\sqrt{3}}{2\sigma_L}kx} - e^{-\frac{\sqrt{3}}{2\sigma_L}ky}|, \tag{3.69}$$

and consequently, for $G_{1,2}(k,t) := e^{-ik^3t}\mathfrak{v}_{1,2}(k)$,

$$\begin{aligned} |v_{1,2}^-(-x,t) - v_{1,2}^-(-y,t)|^2 &= \left| \int_0^\infty \left(e^{-i\frac{\tilde{\alpha}}{\sigma_L}kx} - e^{-i\frac{\tilde{\alpha}}{\sigma_L}ky} \right) e^{-ik^3t}\mathfrak{v}_{1,2}(k)\, dk \right|^2 \\ &\lesssim_{\sigma_L} \int_0^\infty \int_0^\infty \left| e^{-\frac{\sqrt{3}}{2\sigma_L}kx} - e^{-\frac{\sqrt{3}}{2\sigma_L}ky} \right| \left| e^{-\frac{\sqrt{3}}{2\sigma_L}\xi x} - e^{-\frac{\sqrt{3}}{2\sigma_L}\xi y} \right| |G_{1,2}(\xi,t)||G_{1,2}(k,t)|\, dk\, d\xi. \end{aligned} \tag{3.70}$$

Since (3.70) is similar to (3.49), we apply the same process (3.50), (3.57) that

$$\left[v_{1,2}^-(\cdot,t)\right]_{\beta-}^2 \lesssim_{\beta,\sigma_L} \int_0^\infty \left(\int_0^\infty e^{-kx} \left((1+|k|^2)^{\frac{\beta}{2}} |G_{1,2}(k,t)| \right) dk \right)^2 dx \lesssim_{\beta,\sigma_L} \|u_0^L\|_{H^\beta(\mathbb{R}^-)}^2, \tag{3.71}$$

where the Lemma 3.4 is used in the last inequality.
For the case $\beta = 0$, exchanging the function $\tilde{G}_{1,1}^j(k,t)$ for $\tilde{G}_{1,2}^j(k,t)$, we find

$$\|\partial_x^j v_{1,2}^-(\cdot,t)\|_{L^2(\mathbb{R}^-)}^2 \lesssim_{\sigma_L,j} \|u_0^L\|_{H^j(\mathbb{R}^-)}^2, \tag{3.72}$$

where

$$\tilde{G}_{1,2}^j(k,t) := e^{-ik^3t}\left(i\frac{\tilde{\alpha}}{\sigma_L}k\right)^j \mathfrak{v}_{1,2}(k). \tag{3.73}$$

In the case $\lfloor s \rfloor \neq 0$ and $\beta \neq 0$. Since α/σ_L and $-\tilde{\alpha}/\sigma_L$ have the same imaginary part, this shows that the function $\partial_x^{\lfloor s \rfloor} v_{1,2}^-(x,t)$ has the same upper-bound as in (3.68). Combining the estimates (3.71) and (3.72), we find

$$\left\| v_{1,2}^-(\cdot,t) \right\|_{H^s(\mathbb{R}^-)} \lesssim_{s,\sigma_L} \|u_0^L\|_{H^s(\mathbb{R}^-)}, \qquad s \geq 0. \tag{3.74}$$

Therefore, we show that for $s \geq 0$,

$$\sup_{t\in[0,T]} \|v_1^-(\cdot,t)\|_{H^s(\mathbb{R}^-)} \leq \sup_{t\in[0,T]} \|v_{1,1}^-(\cdot,t)\|_{H^s(\mathbb{R}^-)} + \sup_{t\in[0,T]} \|v_{1,2}^-(\cdot,t)\|_{H^s(\mathbb{R}^-)} \lesssim_{s,\sigma_L} \|u_0^L\|_{H^s(\mathbb{R}^-)}. \tag{3.75}$$

□

Proof of (3.45a). As before, we only focus on the estimates of the functions $v_{1,1}^-(x,t)$ and $v_{1,2}^-(x,t)$, the proofs for the other functions $v_j^\pm(x,t)$ are analogous.

To prove the estimate, we first define a smooth cut-off function $\theta_2(k)$ and a function $V_{1,1}(x,t)$ as follows:

$$\theta_2(k) = \begin{cases} 1, & |k| \leq 1, \\ 0, & |k| \geq 2, \end{cases} \tag{3.76}$$

and

$$V_{1,1}^-(x,t) = \int_0^\infty e^{-i\frac{\alpha}{\sigma_L}kx+ik^3t}\alpha\widehat{U_0^L}\left(\frac{-\alpha^2k}{\sigma_L}\right)dk := \mathrm{II}_1^{(V)}(x,t) + \mathrm{II}_2^{(V)}(x,t), \tag{3.77}$$

where

$$\mathrm{II}_1^{(V)}(x,t) := \int_0^\infty e^{-i\frac{\alpha}{\sigma_L}kx+ik^3t}\theta_2(k)\alpha\widehat{U_0^L}\left(\frac{-\alpha^2k}{\sigma_L}\right)dk, \quad \mathrm{II}_2^{(V)}(x,t) := \int_0^\infty e^{-i\frac{\alpha}{\sigma_L}kx+ik^3t}(1-\theta_2(k))\alpha\widehat{U_0^L}\left(\frac{-\alpha^2k}{\sigma_L}\right)dk. \tag{3.78}$$

We may write $\partial_t^j \mathrm{II}_1^{(V)}(x,t)$ in the following way:

$$\partial_t^j \mathrm{II}_1^{(V)}(x,t) = \int_0^2 \left[e^{-i\frac{\alpha}{\sigma_L}kx+ik^3t}\theta_2(k)\alpha(ik^3)^j(1+k^2)^{-\frac{s}{2}}\right]\left[(1+k^2)^{\frac{s}{2}}\widehat{U_0^L}\left(\frac{-\alpha^2k}{\sigma_L}\right)\right]dk, \qquad j = 0,1,2,\ldots. \tag{3.79}$$

Note that $\alpha = e^{i\frac{2\pi}{3}}$ and $\sigma_L > 0$, we have

$$|e^{-i\frac{\alpha}{\sigma_L}kx}| \leq 1, \qquad x < 0. \tag{3.80}$$

Applying Hölder inequality, we obtain

$$|\partial_t^j \mathrm{II}_1^{(V)}(x,t)| \leq C\left(\int_0^2 \frac{|k|^{6j}}{(1+k^2)^s}dk\right)^{\frac{1}{2}} \|U_0^L\|_{H^s(\mathbb{R})} \lesssim_{s,\sigma_L,j} \|u_0^L\|_{H^s(\mathbb{R}^-)}, \tag{3.81}$$

and hence,

$$\|\partial_t^j \mathrm{II}_1^{(V)}(x,\cdot)\|_{L^2((0,T))} \lesssim_{s,\sigma_L,j} T^{\frac{1}{2}}\|u_0^L\|_{H^s(\mathbb{R}^-)}. \tag{3.82}$$

For the noninteger case and $0 < m_0 < 1$, we have

$$\|\mathrm{II}_1^{(V)}(x,\cdot)\|_{H^{m_0}((0,T))} \leq \|\mathrm{II}_1^{(V)}(x,\cdot)\|_{L^2((0,T))}^{1-m_0}\|\mathrm{II}_1^{(V)}(x,\cdot)\|_{H^1((0,T))}^{m_0} \lesssim_{s,\sigma_L} T^{\frac{1}{2}}\|u_0^L\|_{H^s(\mathbb{R}^-)}. \tag{3.83}$$

On the other hand, the second term $\mathrm{II}_2^{(V)}(x,t)$ can be written as follows:

$$\mathrm{II}_2^{(V)}(x,t) = \int_1^\infty e^{-i\frac{\alpha}{\sigma_L}kx+ik^3t}(1-\theta_2(k))\alpha\widehat{U_0^L}\left(\frac{-\alpha^2k}{\sigma_L}\right)dk = \int_1^\infty e^{i\tau t}\left[e^{-i\frac{\alpha}{\sigma_L}\tau^{\frac{1}{3}}x}\left(1-\theta_2\left(\tau^{\frac{1}{3}}\right)\right)\alpha\widehat{U_0^L}\left(\frac{-\alpha^2\tau^{\frac{1}{3}}}{\sigma_L}\right)\frac{\tau^{\frac{-2}{3}}}{3}\right]d\tau, \tag{3.84}$$

where the change of variable $k = \tau^{1/3}$ is used. Thus, we find

$$\|\mathrm{II}_2^{(V)}(x,\cdot)\|_{H^{m_0}((0,T))}^2 \leq C\int_1^\infty |\tau|^{2m_0-\frac{4}{3}}\left|\widehat{U_0^L}\left(\frac{-\alpha^2\tau^{\frac{1}{3}}}{\sigma_L}\right)\right|^2 d\tau \lesssim_{s,\sigma_L} \|U_0^L\|_{H^s(\mathbb{R})}^2 \lesssim \|u_0^L\|_{H^s(\mathbb{R}^-)}^2. \tag{3.85}$$

Therefore, for $0 < m_0 < 1$, we have

$$\sup_{x\in\mathbb{R}^-} \|v_{1,1}^-(x,\cdot)\|_{H^{m_0}((0,T))} \le \sup_{x\in\mathbb{R}^-} \|\mathrm{II}_1^{(V)}(x,\cdot)\|_{H^{m_0}((0,T))} + \sup_{x\in\mathbb{R}^-} \|\mathrm{II}_2^{(V)}(x,\cdot)\|_{H^{m_0}((0,T))} \lesssim_{s,\sigma_L} (1+T^{\frac{1}{2}})\|u_0^L\|_{H^s(\mathbb{R}^-)}. \tag{3.86}$$

Next, we examine the function $v_{1,2}^-(x,t)$. Let

$$V_{1,2}^-(x,t) = \int_0^\infty e^{i\frac{\tilde{\alpha}}{\sigma_L}kx - ik^3 t}\theta_2(k)\tilde{\alpha}\widehat{U_0^L}\left(\frac{k}{\sigma_L}\right)dk + \int_0^\infty e^{i\frac{\tilde{\alpha}}{\sigma_L}kx - ik^3 t}(1-\theta_2(k))\tilde{\alpha}\widehat{U_0^L}\left(\frac{k}{\sigma_L}\right)dk := \mathrm{II}_1^{(\tilde{V})}(x,t) + \mathrm{II}_2^{(\tilde{V})}(x,t). \tag{3.87}$$

Applying the Sobolev interpolation and Hölder inequality again, we find that for $0 < m_0 < 1$,

$$\|\mathrm{II}_1^{(\tilde{V})}(x,\cdot)\|_{H^{m_0}((0,T))} \lesssim_{s,\sigma_L} T^{\frac{1}{2}}\|u_0^L\|_{H^s(\mathbb{R}^-)}, \tag{3.88}$$

where the following fact is used

$$|e^{i\frac{\tilde{\alpha}}{\sigma_L}kx}| \le 1, \qquad x < 0. \tag{3.89}$$

Moreover, by the change of variable $k = (-\tau)^{1/3}$, we find

$$\mathrm{II}_2^{(\tilde{V})}(x,t) = \int_{-\infty}^{-1} e^{i\tau t}\left[e^{i\frac{\tilde{\alpha}}{\sigma_L}(-\tau)^{\frac{1}{3}}x}\left(1-\theta_2\left((-\tau)^{\frac{1}{3}}\right)\right)\tilde{\alpha}\widehat{U_0^L}\left(\frac{(-\tau)^{\frac{1}{3}}}{\sigma_L}\right)\frac{(-\tau)^{\frac{-2}{3}}}{3}\right]d\tau. \tag{3.90}$$

This yields the estimate

$$\|\mathrm{II}_2^{(\tilde{V})}(x,\cdot)\|_{H^{m_0}((0,T))}^2 \lesssim_{s,\sigma_L} \|U_0^L\|_{H^s(\mathbb{R})}^2 \lesssim \|u_0^L\|_{H^s(\mathbb{R}^-)}^2. \tag{3.91}$$

Combining these two inequalities (3.88) and (3.91), we conclude that

$$\sup_{x\in\mathbb{R}^-} \|v_{1,2}^-(x,\cdot)\|_{H^{m_0}((0,T))} \le \sup_{x\in\mathbb{R}^-} \|\mathrm{II}_1^{(\tilde{V})}(x,\cdot)\|_{H^{m_0}((0,T))} + \sup_{x\in\mathbb{R}^-} \|\mathrm{II}_2^{(\tilde{V})}(x,\cdot)\|_{H^{m_0}((0,T))} \lesssim_{s,\sigma_L} (1+T^{\frac{1}{2}})\|u_0^L\|_{H^s(\mathbb{R}^-)}. \tag{3.92}$$

□

Proof of (3.45b). The proof of (3.45b) is similar to that of (3.45a). The only difference is that, when considering the functions $(v_{1,1}^-)_x$ and $(v_{1,2}^-)_x$, the Sobolev exponent used becomes $s/3$. With this modification, the estimate (3.45b) follows the same argument as in the proof of (3.45a). □

3.3 | The Estimates of the Functions (3.7a) and (3.7b)

In this subsection, we establish the following estimates for the functions $w_j^\pm(x,t)$:

$$\sup_{t\in[0,T]} \|w_j^\pm(\cdot,t)\|_{H^s(\mathbb{R}^\pm)} \lesssim_{s,\sigma_L,\sigma_R} \int_0^T \left(\|f_L(\cdot,t)\|_{H^s(\mathbb{R}^-)} + \|f_R(\cdot,t)\|_{H^s(\mathbb{R}^+)}\right)dt, \qquad s \ge 0. \tag{3.93a}$$

Moreover, we derive the following time estimates:

$$\sup_{x\in\mathbb{R}^\pm} \|w_j^\pm(x,\cdot)\|_{H^{(s+1)/3}((0,T))} \lesssim_{s,\sigma_L,\sigma_R} (1+T^{\frac{1}{2}})\left(\|f_L\|_{L^2([0,T];H^s(\mathbb{R}^-))} + \|f_R\|_{L^2([0,T];H^s(\mathbb{R}^+))}\right), \qquad -1 \le s < 2, \tag{3.94a}$$

$$\sup_{x\in\mathbb{R}^\pm} \|(w_j^\pm)_x(x,\cdot)\|_{H^{s/3}((0,T))} \lesssim_{s,\sigma_L,\sigma_R} (1+T^{\frac{1}{2}})\left(\|f_L\|_{L^2([0,T];H^s(\mathbb{R}^-))} + \|f_R\|_{L^2([0,T];H^s(\mathbb{R}^+))}\right), \qquad 0 \le s < 3. \tag{3.94b}$$

Proof of (3.93a). As before, we only focus on the estimates of the functions $w^-_{1,1}(x,t)$ and $w^-_{1,2}(x,t)$. The proofs for the other functions $w^\pm_j(x,t)$ are analogous.

Applying Minkowski's integral inequality, we obtain

$$\left\|w^-_{1,j}(\cdot,t)\right\|_{H^s(\mathbb{R}^-)} \le \int_0^T \|\nu^-_{1,j}(\cdot,t-\tau)\|_{H^s(\mathbb{R}^-)}\,d\tau, \qquad j=1,2, \tag{3.95}$$

where the function $\nu^-_{1,j}(x,t-\tau)$ are denoted by

$$\nu^-_{1,1}(x,t-\tau) := \int_0^\infty e^{-i\frac{\alpha}{\sigma_L}kx+ik^3t}\left(e^{-ik^3\tau}\mathfrak{F}_{1,1}(k,\tau)\right)dk, \tag{3.96a}$$

$$\nu^-_{1,2}(x,t-\tau) := \int_0^\infty e^{i\frac{\tilde{\alpha}}{\sigma_L}kx-ik^3t}\left(e^{-ik^3\tau}\mathfrak{F}_{1,2}(k,\tau)\right)dk. \tag{3.96b}$$

Thus, applying the estimate (3.44a) to $\nu^-_{1,1}(x,t-\tau)$ and $\nu^-_{1,2}(x,t-\tau)$, we find that for $s\ge 0$,

$$\sup_{t\in[0,T]}\|w^-_1(\cdot,t)\|_{H^s(\mathbb{R}^-)} \le \sup_{t\in[0,T]}\|w^-_{1,1}(\cdot,t)\|_{H^s(\mathbb{R}^-)} + \sup_{t\in[0,T]}\|w^-_{1,2}(\cdot,t)\|_{H^s(\mathbb{R}^-)} \lesssim_{s,\sigma_L} \int_0^T \|f_L(\cdot,\tau)\|_{H^s(\mathbb{R}^-)}d\tau. \tag{3.97}$$

□

Proof of (3.94a). As before, we only focus on the functions $w^-_{1,1}(x,t)$ and $w^-_{1,2}(x,t)$. To establish the estimate (3.94a), we follow the same idea used in the derivation of the estimates in (3.45a), from which the desired results can be obtained. Specifically, we use the cut-off function $\theta_2(k)$ to divide the integral into two parts. The part involving $\theta_2(k)$, which has support in $[-2,2]$, can be estimated using Minkowski's inequality and Hölder inequality. The remaining part, involving the term $1-\theta_2(k)$, can be handled using Minkowski's inequality. Therefore, we show that for $-1\le s<2$,

$$\sup_{x\in\mathbb{R}^-}\|w^-_{1,1}(x,\cdot)\|_{H^{\frac{s+1}{3}}((0,T))} + \sup_{x\in\mathbb{R}^-}\|w^-_{1,2}(x,\cdot)\|_{H^{\frac{s+1}{3}}((0,T))} \lesssim_{s,\sigma_L} (1+T^{\frac{1}{2}})\|f_L\|_{L^2([0,T];H^s(\mathbb{R}^-))}. \tag{3.98}$$

□

Proof of (3.94b). The proof of (3.94b) is similar to that of (3.94a). The only difference is that, when considering the function $(w^-_{1,1})_x$, the Sobolev exponent used becomes $s/3$. With this modification, the estimate (3.94b) follows the same argument as in the proof of (3.94a). □

3.4 | The Estimates of the Functions (3.8b) and (3.8c)

In this subsection, we follow the approach developed in [21, 22, 25] to establish estimates related to the functions $H^\pm_j(x,t)$. Suppose that the assumptions stated in Theorem 1.1 hold, namely, $\mu_0\in H^{(s+1)/3}((0,T))$ and $\mu_1\in H^{s/3}((0,T))$. Then, we have the following estimates:

$$\sup_{t\in[0,T]}\left\|H^\pm_0(\cdot,t)\right\|_{H^s(\mathbb{R}^\pm)} \lesssim_{s,\sigma_L,\sigma_R} \|\mu_0\|_{H^{\frac{s+1}{3}}((0,T))}, \quad \frac{1}{2}<s<2, \tag{3.99a}$$

$$\sup_{t\in[0,T]}\left\|H^\pm_1(\cdot,t)\right\|_{H^s(\mathbb{R}^\pm)} \lesssim_{s,\sigma_L,\sigma_R} \|\mu_1\|_{H^{\frac{s}{3}}((0,T))}, \quad 0\le s<\frac{3}{2}. \tag{3.99b}$$

Moreover, we have

$$\sup_{x\in\mathbb{R}^\pm}\|H^\pm_0(x,\cdot)\|_{H^{(s+1)/3}((0,T))} + \sup_{x\in\mathbb{R}^\pm}\|(H^\pm_0)_x(x,\cdot)\|_{H^{s/3}((0,T))} \lesssim_{s,\sigma_L,\sigma_R} \|\mu_0\|_{H^{(s+1)/3}((0,T))}, \tag{3.100a}$$

where $1/2 < s < 2$, and

$$\sup_{x\in\mathbb{R}^\pm} \|H_1^\pm(x,\cdot)\|_{H^{(s+1)/3}((0,T))} + \sup_{x\in\mathbb{R}^\pm} \|(H_1^\pm)_x(x,\cdot)\|_{H^{s/3}((0,T))} \lesssim_{s,\sigma_L,\sigma_R} (1+T^{\frac{1}{2}})\|\mu_1\|_{H^{s/3}((0,T))}, \tag{3.100b}$$

where $0 \le s < \frac{3}{2}$.

Proof of (3.99). The proof of (3.99) is similar to the proof of the estimates (3.44a), but the difference is that we shall extend the function $\mu_0(t)$ when $(s+1)/3 > 1/2$. As before, we only focus on the estimates of the functions $H_{0,1}^-(x,t)$ and $H_{1,1}^-(x,t)$, the proofs for the other functions are analogous.

Since $u_0^L(0) = u_0^R(0)$ and $\mu_0(t) = u_0^R(t) - u_0^L(t)$, it follows that $\mu_0(0) = 0$. However, $\mu_0(T)$ may not necessarily vanish. Thus, we cannot express $M_0(-ik^3,T)$ as $\widehat{\mu}_0(k^3)$ directly. To overcome this issue, we define a smooth cut-off function $\theta_3(t)$ as follows:

$$\theta_3(t) = \begin{cases} 1, & |t| \le 1, \\ 0, & |t| \ge 2. \end{cases} \tag{3.101}$$

Furthermore, letting $m_0 = (s+1)/3$, we choose an extension function $\mu \in H^{m_0}(\mathbb{R})$ of $\mu_0 \in H^{m_0}((0,T))$ satisfying

$$\|\mu\|_{H^{m_0}(\mathbb{R})} \le C\|\mu_0\|_{H^{m_0}((0,T))}, \qquad \mu(t) = \mu_0(t), \quad \text{for } t\in[0,T]. \tag{3.102}$$

Let $\tilde{\mu}(t) = \theta_3(t)\mu(t)$ and define

$$\mu^*(t) = \begin{cases} \tilde{\mu}(t), & t\in[0,2], \\ 0, & t\notin[0,2]. \end{cases} \tag{3.103}$$

By this extension, we have

$$\|\mu^*\|_{H^{m_0}(\mathbb{R})} \le C\|\tilde{\mu}\|_{H^{m_0}(\mathbb{R})} \le C\|\mu\|_{H^{m_0}(\mathbb{R})} \le C\|\mu_0\|_{H^{m_0}((0,T))}, \qquad m_0 > \frac{1}{2}. \tag{3.104}$$

In addition, when comparing the integrals $v_{1,1}^-(x,t)$ and $H_{0,1}^-(x,t)$, we find that the proof of the estimates (3.99a) is analogous to that of (3.44a); it only requires replacing the function $\mathfrak{v}_{1,1}(k)$ with $k^2 M_0(-ik^3,T)$. As a result, we obtain

$$\left\|H_{0,1}^-(\cdot,t)\right\|_{H^s(\mathbb{R}^-)}^2 \lesssim_{s,\sigma_L} \int_0^\infty |k|^{2s}\left|k^2\int_0^T e^{-ik^3\tau}\mu_0(\tau)\,d\tau\right|^2 dk \le \int_0^\infty |k|^{2s+4}\left|\widehat{\mu^*}(k^3)\right|^2 dk. \tag{3.105}$$

By the change of variables $k = \tau^{1/3}$, we obtain

$$\|H_{0,1}^-(\cdot,t)\|_{H^s(\mathbb{R}^-)} \lesssim_{s,\sigma_L,\theta_3} \|\mu^*\|_{H^{m_0}(\mathbb{R})} \lesssim \|\mu_0\|_{H^{m_0}((0,T))}, \qquad \frac{1}{2} < s < 1. \tag{3.106}$$

To prove the estimates for $H_{1,1}^-(x,t)$, we simply apply the same argument as in (3.44a) to obtain the result (3.99b). Indeed, since $0 \le s < 3/2$ and $\mu_1 \in H^{s/3}((0,T))$, we have $\mu_1(0) = \mu_1(T) = 0$ and thus $M_1(-ik^3,T) = \widehat{\mu}_1(k^3)$. Furthermore, by replacing the function $\mathfrak{v}_{1,1}(k)$ with $-\alpha^2 ik\widehat{\mu}_1(k^3)$ and applying the estimate in (3.44a), we deduce that

$$\|H_{1,1}^-(\cdot,t)\|_{H^s(\mathbb{R}^-)} \lesssim_{s,\sigma_L} \|\mu_1\|_{H^{\frac{s}{3}}((0,T))}. \tag{3.107}$$

□

Proof of (3.100a). By the change of variable $k = \tau^{1/3}$, we have

$$H_{0,1}^-(x,t) = \frac{1}{3}\int_0^\infty e^{i\tau t}\left(e^{-i\frac{\alpha}{\sigma_L}\tau^{\frac{1}{3}}x} M_0(-i\tau,T)\right)d\tau \tag{3.108}$$

which implies that

$$\|H_{0,1}^{-}(x,\cdot)\|_{H^{m_0}((0,T))}^2 \leq C\int_0^\infty (1+\tau^2)^{m_0}|\widehat{\mu^*}(\tau)|^2\,d\tau \leq C\|\mu_0\|_{H^{m_0}((0,T))}^2, \qquad \frac{1}{2}<s<2. \tag{3.109}$$

Moreover, by letting $m_1 := s/3$, we can proceed in the same way to obtain

$$\|(H_{0,1}^{-})_x(x,\cdot)\|_{H^{m_1}((0,T))} \lesssim_{s,\sigma_L} \|\mu_0\|_{H^{m_0}((0,T))}. \tag{3.110}$$

Therefore, the desired estimates for $H_{0,1}^{-}(x,t)$ and its spatial derivative follow. □

Proof of (3.100b). Again, we only focus on the cases for $H_{1,1}^{-}(x,t)$ and $(H_{1,1}^{-})_x(x,t)$. To establish the estimate for $H_{1,1}^{-}(x,t)$, we follow the same approach as in (3.45a). Indeed, when comparing the integrals $v_{1,1}^{-}(x,t)$ and $H_{1,1}^{-}(x,t)$, we find that the proof of the estimate for $H_{1,1}^{-}(x,t)$ is analogous to that of (3.45). The only difference is that the function $\mathfrak{v}_{1,1}(k)$ is replaced by $-i\alpha^2 k\widehat{\mu}_1(k^3)$. Thus, we find

$$\|H_{1,1}^{-}(x,\cdot)\|_{H^{m_0}((0,T))} \lesssim_{s,\sigma_L} \|\mu_1\|_{H^{m_1}((0,T))}, \qquad 0\leq s<\frac{3}{2}. \tag{3.111}$$

Furthermore, the estimate for $(H_{1,1}^{-})_x(x,t)$ can be proved in the same way. It only requires replacing the function $-i\frac{\alpha}{\sigma_L}\mathfrak{v}_{1,1}(k)$ with $-\frac{1}{\sigma_L}k^2\widehat{\mu}_1(k^3)$. Thus, we obtain

$$\|(H_{1,1}^{-})_x(x,\cdot)\|_{H^{m_1}((0,T))} \lesssim_{s,\sigma_L} \|\mu_1\|_{H^{m_1}((0,T))}. \tag{3.112}$$

□

4 | Solution Space and the Proof of Bilinear Estimate

In this section, we derive an estimate for the nonlinear term uu_x, which plays a critical role in constructing the solution space. To derive the inequality in Corollary 1.4, we only need to show the following bilinear estimate:

Lemma 4.1. *Let $0\leq s<1$ and $0<T\leq 1$. Then the following inequality holds:*

$$\int_0^T \|uv_x(\cdot,t)\|_{H^s(\mathbb{R}^\pm)}\,dt \leq C\max\{T^{\frac{1}{2}},T^{\frac{3}{4}}\}\Big(\Lambda_{\pm,3}^T(u)\Lambda_{\pm,4}^T(v)+\Lambda_{\pm,3}^T(u)\Lambda_{\pm,2,s}^T(v)+\Lambda_{\pm,1,s}^T(u)\Lambda_{\pm,4}^T(v)\Big), \tag{4.1}$$

where C is a constant and the Λ-norms are defined in (1.13).

Proof. Using Hölder's inequality, we have

$$\int_0^T \|uv_x(\cdot,t)\|_{H^s(\mathbb{R}^\pm)}\,dt \leq T^{\frac{1}{2}}\left(\int_0^T \|uv_x(\cdot,t)\|_{L^2(\mathbb{R}^\pm)}^2+[uv_x(\cdot,t)]_{s^\pm}^2\,dt\right)^{\frac{1}{2}}. \tag{4.2}$$

For the first term on the right-hand side of (4.2), we find

$$\begin{aligned}\int_0^T \|uv_x(\cdot,t)\|_{L^2(\mathbb{R}^\pm)}^2\,dt &= \int_{\mathbb{R}^\pm}\left(\int_0^T |uv_x(x,t)|^2\,dt\right)dx\\ &\leq \int_0^T \sup_{x\in\mathbb{R}^\pm}|u(x,t)|^2\int_{\mathbb{R}^\pm}|v(x,t)|^2\,dx\,dt\\ &\leq T^{\frac{1}{2}}\Lambda_{\pm,3}^T(u)^2\Lambda_{\pm,4}^T(v)^2.\end{aligned} \tag{4.3}$$

To handle the second term of (4.2), we use the following facts:

$$[f]_{s\pm}^2 = \int_{\mathbb{R}\pm}\int_{\mathbb{R}\pm} \frac{|f(x)-f(y)|^2}{|x-y|^{1+2s}}dxdy = 2\int_{\mathbb{R}\pm}\int_{\mathbb{R}\pm} \frac{|f(x+z)-f(x)|^2}{|z|^{1+2s}}dzdx \tag{4.4}$$

and decompose the term $[uv_x(\cdot,t)]_{s\pm}$ into two components as follows:

$$\int_0^T [uv_x(\cdot,t)]_{s\pm}^2\, dt \le \mathrm{I}_{1,1}^{\pm} + \mathrm{I}_{1,2}^{\pm}, \tag{4.5}$$

where

$$\mathrm{I}_{1,1}^{\pm} := 4\int_0^T \int_{\mathbb{R}\pm}\int_{\mathbb{R}\pm} \frac{|(u(x+z,t)-u(x,t))v_x(x+z,t)|^2}{|z|^{1+2s}}\, dzdxdt, \tag{4.6a}$$

$$\mathrm{I}_{1,2}^{\pm} := 4\int_0^T \int_{\mathbb{R}\pm}\int_{\mathbb{R}\pm} \frac{|u(x,t)(v_x(x+z,t)-v_x(x,t))|^2}{|z|^{1+2s}}\, dzdxdt. \tag{4.6b}$$

In the first integral $\mathrm{I}_{1,1}^{\pm}$, we find that

$$\begin{aligned}
\mathrm{I}_{1,1}^{\pm} &\le 4\int_0^T \sup_{x\in\mathbb{R}\pm} |v(x,t)|^2 \left(\int_{\mathbb{R}\pm}\int_{\mathbb{R}\pm} \frac{|u(x+z,t)-u(x,t)|^2}{|z|^{1+2s}}\, dzdx\right)dt \\
&\le 2\int_0^T \sup_{x\in\mathbb{R}\pm} |v(x,t)|^2 \|u(\cdot,t)\|_{H^s(\mathbb{R}\pm)}^2 dt \\
&\le 2T^{\frac{1}{2}} \Lambda_{\pm,1,s}^T(u)^2 \Lambda_{\pm,4}^T(v)^2.
\end{aligned} \tag{4.7}$$

Next, we consider the estimate for the second integral $\mathrm{I}_{1,2}^{\pm}$. Since

$$\int_0^T \int_{\mathbb{R}\pm} \frac{|v_x(x+z,t)-v_x(x,t)|^2}{|z|^{1+2s}}\, dzdt = \int_0^T |D_x^s v_x(x,t)|^2\, dt, \tag{4.8}$$

it follows that

$$\mathrm{I}_{1,2}^{\pm} \le 4\int_{\mathbb{R}\pm} \sup_{t\in[0,T]} |u(x,t)|^2 \left(\int_0^T \int_{\mathbb{R}\pm} \frac{|v_x(x+z,t)-v_x(x,t)|^2}{|z|^{1+2s}}\, dzdt\right)dx \le 4\Lambda_{\pm,3}^T(u)^2\Lambda_{\pm,2,s}^T(v)^2, \tag{4.9}$$

where Hölder's inequality is used. Combining the inequalities (4.3), (4.5), (4.7), and (4.9), we conclude that

$$\int_0^T \|uv_x(\cdot,t)\|_{H^s(\mathbb{R}\pm)} \le C\max\left\{T^{\frac{1}{2}}, T^{\frac{3}{4}}\right\}\left(\Lambda_{\pm,3}^T(u)\Lambda_{\pm,4}^T(v) + \Lambda_{\pm,1,s}^T(u)\Lambda_{\pm,4}^T(v) + \Lambda_{\pm,3}^T(u)\Lambda_{\pm,2,s}^T(v)\right), \tag{4.10}$$

where $0 \le s < 1$. □

5 | Lambda Norm Estimates of the Linear Forced Solutions

After handling the nonlinear terms, we require a more refined solution space than $C([0,T]; H^s(\mathbb{R}))$, namely, the space X_T. To ensure that the map Φ, defined in (1.25), is a contraction map, we need to further analyze the linear solutions (1.17a) and (1.17b). The precise estimates are presented in Theorem 2.1. Recall that

$$\Lambda_{\pm,1,s}^T(u) = \sup_{t\in[0,T]} \|u(\cdot,t)\|_{H^s(\mathbb{R}\pm)}. \tag{5.1}$$

The estimates for the $\Lambda_{\pm,1,s}^T$ norms in (2.2) have already been established in Section 3. We now proceed to analyze the $\Lambda_{\pm,2,s}^T$ norms.

Theorem 5.1. *Let $1/2 < s < 1$. Suppose that $u_0^L \in H^s(\mathbb{R}^-)$, $u_0^R \in H^s(\mathbb{R}^+)$, $\mu_0 \in H^{(s+1)/3}((0,T))$ and $\mu_1 \in H^{s/3}((0,T))$. Then,*

$$\Lambda^T_{-,2,s}(v^L) + \Lambda^T_{+,2,s}(v^R) \lesssim_{s,\sigma_L,\sigma_R} \|(u_0,\mu_0,\mu_1)\|_D + \int_0^T \left(\|f_L(\cdot,t)\|_{H^s(\mathbb{R}^-)} + \|f_R(\cdot,t)\|_{H^s(\mathbb{R}^+)}\right)dt, \tag{5.2}$$

where $v^L(x,t) = S_L[u_0,\mu_0,\mu_1; f]$ and $v^R(x,t) = S_R[u_0,\mu_0,\mu_1; f]$ are the solutions defined in (1.17).

Proof. To establish the estimates in (5.2), it suffices to prove the following inequality:

$$\Lambda^T_{-,2,s}(\mathcal{U}_L + \mathcal{F}_L + v_1^- + w_1^- + H_0^- + H_1^-) \lesssim_{s,\sigma_L} \|(u_0,\mu_0,\mu_1)\|_D + \int_0^T \|f_L(\cdot,t)\|_{H^s(\mathbb{R}^-)}dt. \tag{5.3}$$

The proof for the others is analogous. Recall that

$$\Lambda^T_{-,2,0}(\mathcal{U}_L) = \sup_{x\in\mathbb{R}^-}\left(\int_0^T \left|\frac{1}{2\pi}\int_{\mathbb{R}} e^{ikx-ik^3\sigma_L^3 t}(ik)\widehat{u_0^L}(k)dk\right|^2 dt\right)^{\frac{1}{2}}. \tag{5.4}$$

We split the integral into two parts:

$$\Lambda^T_{-,2,0}(\mathcal{U}_L) = \sup_{x\in\mathbb{R}^-}\left(\mathcal{I}_1{}^- + \mathcal{I}_1{}^+\right)^{\frac{1}{2}}, \tag{5.5}$$

where

$$\mathcal{I}_1{}^- := \int_0^T \left|\frac{1}{2\pi}\int_{-\infty}^0 e^{ikx-ik^3\sigma_L^3 t}(ik)\widehat{u_0^L}(k)\,dk\right|^2 dt, \qquad \mathcal{I}_1{}^+ := \int_0^T \left|\frac{1}{2\pi}\int_0^\infty e^{ikx-ik^3\sigma_L^3 t}(ik)\widehat{u_0^L}(k)\,dk\right|^2 dt. \tag{5.6}$$

Using the change of variable $k = -\tau^{\frac{1}{3}}/\sigma_L$ to the integral $\mathcal{I}_1{}^-$ and applying Plancherel's theorem, we obtain

$$\mathcal{I}_1{}^- = C\int_0^T \left|\frac{1}{2\pi}\int_0^\infty e^{i\tau t}\left(e^{-i\frac{\tau^{\frac{1}{3}}}{\sigma_L}x}\tau^{-\frac{1}{3}}\widehat{u_0^L}\left(\frac{-\tau^{\frac{1}{3}}}{\sigma_L}\right)\right)d\tau\right|^2 dt \lesssim_{\sigma_L} \int_0^\infty |\tau|^{-\frac{2}{3}}\left|\widehat{u_0^L}\left(\frac{-\tau^{\frac{1}{3}}}{\sigma_L}\right)\right|^2 d\tau \lesssim_{\sigma_L} \|u_0^L\|^2_{L^2(\mathbb{R}^-)}, \tag{5.7}$$

where we have used the change of variable $\tau = -\sigma_L^3 k^3$ in the last inequality. Similarly, using the change of variables $k = (-\tau)^{\frac{1}{3}}/\sigma_L$ and applying Plancherel's theorem to $\mathcal{I}_1{}^+$, we obtain

$$\mathcal{I}_1{}^+ \lesssim_{\sigma_L} \int_{-\infty}^0 |\tau|^{-\frac{2}{3}}\left|\widehat{u_0^L}\left(\frac{(-\tau)^{\frac{1}{3}}}{\sigma_L}\right)\right|^2 d\tau \lesssim_{\sigma_L} \|U_0^L\|^2_{L^2(\mathbb{R})} \lesssim \|u_0^L\|^2_{L^2(\mathbb{R}^-)}. \tag{5.8}$$

Therefore, for $s = 0$, we have

$$\Lambda^T_{-,2,0}(\mathcal{U}_L) \lesssim_{\sigma_L} \|u_0^L\|_{L^2(\mathbb{R}^-)}. \tag{5.9}$$

For the case where $0 < s < 1$, we write

$$\Lambda^T_{-,2,s}(\mathcal{U}_L) = \sup_{x\in\mathbb{R}^-}\left(\mathcal{I}_2{}^- + \mathcal{I}_2{}^+\right)^{\frac{1}{2}}, \tag{5.10}$$

where

$$\mathcal{I}_2{}^- := \int_0^T\int_{-\infty}^0 \frac{1}{|z|^{1+2s}}\left|\frac{1}{2\pi}\int_{-\infty}^0 e^{ikx-ik^3\sigma_L^3 t}(ik)\widehat{u_0^L}(k)\,(e^{ikz}-1)dk\right|^2 dzdt, \tag{5.11a}$$

$$\mathcal{I}_2{}^{+} := \int_0^T \int_{-\infty}^0 \frac{1}{|z|^{1+2s}} \left| \frac{1}{2\pi} \int_0^\infty e^{ikx - ik^3\sigma_L^3 t} (ik) \widehat{u_0^L}(k) (e^{ikz} - 1) dk \right|^2 dz dt. \tag{5.11b}$$

Following the same approach as in $\mathcal{I}_1{}^{-}$ and $\mathcal{I}_1{}^{+}$, we obtain

$$\mathcal{I}_2{}^{-} \leq C \int_{-\infty}^0 \frac{1}{|z|^{1+2s}} \left(\int_0^\infty |\tau|^{-\frac{2}{3}} \left| \widehat{u_0^L}\left(\frac{-\tau^{\frac{1}{3}}}{\sigma_L} \right) \right|^2 |e^{-i\frac{\tau^{\frac{1}{3}}}{\sigma_L} z} - 1|^2 d\tau \right) dz = C \int_{-\infty}^0 |\widehat{u_0^L}(k)|^2 \left(\int_{-\infty}^0 \frac{|e^{ikz} - 1|^2}{|z|^{1+2s}} dz \right) dk, \tag{5.12a}$$

and

$$\mathcal{I}_2{}^{+} \leq C \int_{-\infty}^0 \frac{1}{|z|^{1+2s}} \left(\int_{-\infty}^0 |\tau|^{-\frac{2}{3}} \left| \widehat{u_0^L}\left(\frac{(-\tau)^{\frac{1}{3}}}{\sigma_L} \right) \right|^2 |e^{i\frac{(-\tau)^{\frac{1}{3}}}{\sigma_L} z} - 1|^2 d\tau \right) dz = C \int_0^\infty |\widehat{u_0^L}(k)|^2 \left(\int_{-\infty}^0 \frac{|e^{ikz} - 1|^2}{|z|^{1+2s}} dz \right) dk, \tag{5.12b}$$

where the change of variables $k = -\tau^{1/3}/\sigma_L$ is used in $\mathcal{I}_2{}^{-}$ and $k = (-\tau)^{1/3}/\sigma_L$ is used in $\mathcal{I}_2{}^{+}$.
Note that

$$\int_{-\infty}^0 \frac{|e^{ikz} - 1|^2}{|z|^{1+2s}} dz = k^{2s} \int_0^\infty \frac{1}{z^{1+2s}} (2 - 2\cos z)\, dz, \tag{5.13a}$$

and

$$\int_0^\infty \frac{1}{z^{1+2s}} (2 - 2\cos z)\, dz < \infty, \qquad \text{for } 0 < s < 1. \tag{5.13b}$$

Therefore, we conclude that

$$\Lambda^T_{-,2,s}(\mathcal{U}_L) \lesssim_{s,\sigma_L} \|U_0^L\|_{H^s(\mathbb{R})} \lesssim \|u_0^L\|_{H^s(\mathbb{R}^-)}. \tag{5.14}$$

To find the upper bound of $\Lambda^T_{-,2,s}(\mathcal{F}_L)$, we note that the function $\mathcal{F}_L(x,t)$ is analogous to $\mathcal{U}_L(x,t)$. The primary difference lies in their definitions (3.2a) and (3.2b), where $\widehat{u_0^L}(k)$ in $\mathcal{U}_L(x,t)$ is replaced by $F_L(k,t)$ in $\mathcal{F}_L(x,t)$. Thus, using Minkowski's integral inequality and the estimate similar to (5.9) and (5.14), we obtain

$$\Lambda^T_{-,2,s}(\mathcal{F}_L) \lesssim_{s,\sigma_L} \int_0^T \|f_L(\cdot,t)\|_{H^s(\mathbb{R}^-)} dt, \qquad s \in [0,1). \tag{5.15}$$

Next, we turn our attention to the estimate of $v^-_{1,1}(x,t)$. By the change of variables $k = \tau^{1/3}$, we find

$$\Lambda^T_{-,2,0}(v^-_{1,1}) = C \sup_{x\in\mathbb{R}^-} \left(\int_0^T \left| \int_0^\infty e^{i\tau t} \left(e^{-i\frac{\alpha}{\sigma_L}\tau^{\frac{1}{3}} x} \tau^{-\frac{1}{3}} \widehat{u_0^L}\left(\frac{-\alpha^2\tau^{\frac{1}{3}}}{\sigma_L} \right) \right) d\tau \right|^2 dt \right)^{\frac{1}{2}}. \tag{5.16}$$

Applying Plancherel's theorem, we obtain the estimate

$$\Lambda^T_{-,2,0}(v^-_{1,1}) \lesssim_{\sigma_L} \sup_{x\in\mathbb{R}^-} \left(\int_0^\infty e^{\frac{\sqrt{3}}{\sigma_L}\tau^{\frac{1}{3}} x} |\tau|^{\frac{-2}{3}} \left| \widehat{u_0^L}\left(\frac{-\alpha^2\tau^{\frac{1}{3}}}{\sigma_L} \right) \right|^2 d\tau \right)^{\frac{1}{2}} \lesssim_{\sigma_L} \|u_0^L\|_{L^2(\mathbb{R}^-)}. \tag{5.17}$$

For the case where $0 < s < 1$, using the same change of variables $k = \tau^{1/3}$ and applying Plancherel's theorem, we obtain

$$\Lambda^T_{-,2,s}(v^-_{1,1}) = C \sup_{x\in\mathbb{R}^-}\left(\int_{-\infty}^0 \frac{1}{|z|^{1+2s}}\left(\int_0^T\left|\int_0^\infty e^{i\tau t}\left(e^{-i\frac{\alpha\tau^{\frac{1}{3}}(x+z)}{\sigma_L}} - e^{-i\frac{\alpha\tau^{\frac{1}{3}}x}{\sigma_L}}\right)\tau^{\frac{-1}{3}}\widehat{u^L_0}\left(\frac{-\alpha^2\tau^{\frac{1}{3}}}{\sigma_L}\right)d\tau\right|^2 dt\right)dz\right)^{\frac{1}{2}}$$

$$\lesssim_{\sigma_L} \sup_{x\in\mathbb{R}^-}\left(\int_{-\infty}^0 \frac{1}{|z|^{1+2s}}\left(\int_0^\infty\left|e^{-i\frac{\alpha\tau^{\frac{1}{3}}(x+z)}{\sigma_L}} - e^{-i\frac{\alpha\tau^{\frac{1}{3}}x}{\sigma_L}}\right|^2|\tau|^{\frac{-2}{3}}\left|\widehat{u^L_0}\left(\frac{-\alpha^2\tau^{\frac{1}{3}}}{\sigma_L}\right)\right|^2 d\tau\right)dz\right)^{\frac{1}{2}}. \tag{5.18}$$

By Lemma 3.2, the following estimate holds:

$$\left|e^{-i\frac{\alpha}{\sigma_L}\tau^{\frac{1}{3}}(x+z)} - e^{-i\frac{\alpha}{\sigma_L}\tau^{\frac{1}{3}}x}\right|^2 \lesssim_{\sigma_L} \left|e^{\frac{\sqrt{3}}{2\sigma_L}\tau^{\frac{1}{3}}(x+z)} - e^{\frac{\sqrt{3}}{2\sigma_L}\tau^{\frac{1}{3}}x}\right|^2. \tag{5.19}$$

Hence, we deduce that

$$\Lambda^T_{-,2,s}(v^-_{1,1}) \lesssim_{\sigma_L} \sup_{x\in\mathbb{R}^-}\left(\int_{-\infty}^0 \frac{1}{|z|^{1+2s}}\left(\int_0^\infty\left|e^{\frac{\sqrt{3}}{2\sigma_L}\tau^{\frac{1}{3}}(x+z)} - e^{\frac{\sqrt{3}}{2\sigma_L}\tau^{\frac{1}{3}}x}\right|^2|\tau|^{\frac{-2}{3}}\left|\widehat{u^L_0}\left(\frac{-\alpha^2\tau^{\frac{1}{3}}}{\sigma_L}\right)\right|^2 d\tau\right)dz\right)^{\frac{1}{2}}$$

$$\lesssim_{\sigma_L} \sup_{x\in\mathbb{R}^-}\left(\int_0^\infty|\tau|^{\frac{-2}{3}}\left|\widehat{u^L_0}\left(\frac{-\alpha^2\tau^{\frac{1}{3}}}{\sigma_L}\right)\right|^2\int_{-\infty}^0\frac{|e^{\frac{\sqrt{3}}{2\sigma_L}\tau^{\frac{1}{3}}(x+z)} - e^{\frac{\sqrt{3}}{2\sigma_L}\tau^{\frac{1}{3}}x}|^2}{|z|^{1+2s}}dz d\tau\right)^{\frac{1}{2}}. \tag{5.20}$$

Note that

$$\int_{-\infty}^0\frac{|e^{\frac{\sqrt{3}}{2\sigma_L}\tau^{\frac{1}{3}}(x+z)} - e^{\frac{\sqrt{3}}{2\sigma_L}\tau^{\frac{1}{3}}x}|^2}{|z|^{1+2s}}dz = Ce^{\frac{\sqrt{3}}{\sigma_L}\tau^{\frac{1}{3}}x}\tau^{\frac{2s}{3}}\int_0^\infty\frac{(e^{-y}-1)^2}{y^{1+2s}}dy \tag{5.21}$$

and

$$\int_0^\infty\frac{(e^{-y}-1)^2}{y^{1+2s}}dy = \begin{cases}4^s(1-2^{1-2s})\Gamma(-2s), & 0<s<1,\ s\neq\frac{1}{2},\\ (\frac{4}{3})^s\sqrt{2\sqrt{3}\ln 2}, & s=\frac{1}{2}.\end{cases} \tag{5.22}$$

We thus conclude that

$$\Lambda^T_{-,2,s}(v^-_{1,1}) \lesssim_{s,\sigma_L} \|u^L_0\|_{H^s(\mathbb{R}^-)}. \tag{5.23}$$

In addition, using estimates similar to those in (5.17), (5.21), and (5.23), we can also derive the estimate for the function $v^-_{1,2}(x,t)$ as follows:

$$\Lambda^T_{-,2,s}(v^-_{1,2}) \lesssim_{s,\sigma_L} \|u^L_0\|_{H^s(\mathbb{R}^-)}, \qquad 0\le s<1. \tag{5.24}$$

Moreover, by applying Minkowski's integral inequality to the functions $w^-_{1,1}(x,t)$ and $w^-_{1,2}(x,t)$, we obtain

$$\Lambda^T_{-,2,s}(w^-_1) \le \Lambda^T_{-,2,s}(w^-_{1,1}) + \Lambda^T_{-,2,s}(w^-_{1,2}) \lesssim_{s,\sigma_L,\sigma_R}\int_0^T\|f_L(\cdot,\tau)\|_{H^s(\mathbb{R}^-)}d\tau, \qquad 0\le s<1. \tag{5.25}$$

Finally, it remains to estimate the functions $H_0^-(x,t)$ and $H_1^-(x,t)$. For the case $1/2 < s < 1$, we follow the approach in (5.18) and apply the inequality (5.19) to derive

$$\Lambda^T_{-,2,s}(H^-_{0,1}) \lesssim_{\sigma_L} \sup_{x\in\mathbb{R}^-}\left(\int_{-\infty}^0 \frac{1}{|z|^{1+2s}}\int_0^\infty \left|e^{-i\frac{\alpha\tau^{\frac{1}{3}}(x+z)}{\sigma_L}} - e^{-i\frac{\alpha\tau^{\frac{1}{3}}x}{\sigma_L}}\right|^2 |\tau|^{\frac{2}{3}}|M_0(-i\tau,T)|^2\,d\tau dz\right)^{\frac{1}{2}}$$

$$\lesssim_{\sigma_L} \sup_{x\in\mathbb{R}^-}\left(\int_0^\infty |\tau|^{\frac{2}{3}}|\widehat{\mu^*}(\tau)|^2\left(\int_{-\infty}^0 \frac{|e^{\frac{\sqrt{3}}{2\sigma_L}\tau^{\frac{1}{3}}(x+z)} - e^{\frac{\sqrt{3}}{2\sigma_L}\tau^{\frac{1}{3}}x}|^2}{|z|^{1+2s}}dz\right)d\tau\right)^{\frac{1}{2}}$$

$$\lesssim_{s,\sigma_L} \|\mu_0\|_{H^{\frac{s+1}{3}}((0,T))}. \tag{5.26}$$

Using estimates similar to those in (5.26), we obtain

$$\Lambda^T_{-,2,s}(H^-_{0,2}) \lesssim_{s,\sigma_L} \|\mu_0\|_{H^{\frac{s+1}{3}}((0,T))}, \qquad \frac{1}{2} < s < 1. \tag{5.27}$$

To estimate $H^-_{1,1}(x,t)$ and $H^-_{1,2}(x,t)$, we apply the same argument (5.17), (5.23) and (5.24) to obtain

$$\Lambda^T_{-,2,s}(H^-_{1,1}) + \Lambda^T_{-,2,s}(H^-_{1,2}) \lesssim_{s,\sigma_L} \|\mu_1\|_{H^{\frac{s}{3}}((0,T))}, \qquad 0 \le s < 1. \tag{5.28}$$

□

Theorem 5.2. *Let $3/4 < s < 1$. Suppose that $u_0^L \in H^s(\mathbb{R}^-)$, $u_0^R \in H^s(\mathbb{R}^+)$, $\mu_0 \in H^{(s+1)/3}((0,T))$, and $\mu_1 \in H^{s/3}((0,T))$. We obtain the following result:*

$$\Lambda^T_{-,3}(v^L) + \Lambda^T_{+,3}(v^R) \lesssim_{s,\sigma_L,\sigma_R} \|(u_0,\mu_0,\mu_1)\|_D + \int_0^T \left(\|f_L(\cdot,t)\|_{H^s(\mathbb{R}^-)} + \|f_R(\cdot,t)\|_{H^s(\mathbb{R}^+)}\right)dt, \tag{5.29}$$

where $v^L(x,t) = S_L[u_0,\mu_0,\mu_1;f]$ and $v^R(x,t) = S_R[u_0,\mu_0,\mu_1;f]$ are the solutions defined in (1.17).

Proof. To establish the estimate in (5.29), it suffices to prove the following inequality:

$$\Lambda^T_{-,3}(\mathcal{U}_L + \mathcal{F}_L + v_1^- + w_1^- + H_0^- + H_1^-) \lesssim_{s,\sigma_L} \|(u_0,\mu_0,\mu_1)\|_D + \int_0^T \|f_L(\cdot,t)\|_{H^s(\mathbb{R}^-)}dt. \tag{5.30}$$

The proof for the others is analogous.

Let

$$\Lambda_3^T(u) := \left(\int_{\mathbb{R}} \sup_{t\in[0,T]} |u(x,t)|^2\,dx\right)^{\frac{1}{2}}. \tag{5.31}$$

We begin by considering the following linear problem on the whole line:

$$\begin{cases} u_t = \sigma_L^3 u_{xxx}, & x\in\mathbb{R},\ t\in(0,T), \\ u(x,0) = U_0^L, \end{cases} \tag{5.32}$$

where U_0^L is the extension function of u_0^L defined in (3.10). The solution of Equation (5.32) is expressed as

$$S(t)U_0^L(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{ikx - ik^3\sigma_L^3 t}\widehat{U_0^L}(k)\,dk. \tag{5.33}$$

We next recall the following result, which provides an estimate for $S(t)U_0^L(x)$.

Lemma 5.3 (Corollary 2.9 in [10]). *Let $s > 3/4$. Then,*

$$\left(\int_{\mathbb{R}} \sup_{|t|\le 1} |S(t)v_0|^2\,dx\right)^{\frac{1}{2}} \le C\|v_0\|_{H^s(\mathbb{R})}, \tag{5.34}$$

where

$$S(t)v_0(x) := \frac{1}{2\pi}\int_{\mathbb{R}} e^{ikx-ik^3t}\widehat{v}_0(k)\,dk \tag{5.35}$$

is the solution to the equation

$$\begin{cases} \partial_t u - \partial_x^3 u = 0, & x, t \in \mathbb{R}, \\ u(x,0) = v_0(x). \end{cases} \tag{5.36}$$

Although the solution (5.35) in this estimate differs from our case (5.33) by a constant factor σ_L^3, this does not affect the estimate. Applying the estimate in Lemma 5.3, we obtain

$$\Lambda_{-,3}^T(\mathcal{U}_L) \le \Lambda_3^T(S(t)U_0^L) \lesssim \|U_0^L\|_{H^s(\mathbb{R})} \le \|u_0^L\|_{H^s(\mathbb{R}^-)}, \qquad \frac{3}{4} < s < 1. \tag{5.37}$$

Next, we estimate $\mathcal{F}_L(x,t)$. By applying Minkowski's integral inequality and using the same estimate as in (5.37), we obtain

$$\Lambda_{-,3}^T(\mathcal{F}_L) \lesssim \int_0^T \|f_L(\cdot,t)\|_{H^s(\mathbb{R}^-)}dt, \qquad \frac{3}{4} < s < 1. \tag{5.38}$$

We now turn to the estimate of $v_{1,1}^-(x,t)$. Note that

$$\Lambda_{-,3}^T(v_{1,1}^-) \le C\left(\int_{-\infty}^0 \left(\int_0^\infty e^{\frac{\sqrt{3}}{2\sigma_L}kx}\left|\widehat{u_0^L}\left(\frac{-\alpha^2 k}{\sigma_L}\right)\right|dk\right)^2 dx\right)^{\frac{1}{2}}. \tag{5.39}$$

Using the change of variable $y = -\sqrt{3}x/(2\sigma_L)$ and Lemma 3.4, we obtain

$$\Lambda_{-,3}^T(v_{1,1}^-) \lesssim_{\sigma_L} \|U_0^L\|_{L^2(\mathbb{R})} \le \|U_0^L\|_{H^s(\mathbb{R})} \lesssim \|u_0^L\|_{H^s(\mathbb{R}^-)}, \qquad 0 \le s < 1. \tag{5.40}$$

In a similar manner, we derive the following estimate:

$$\Lambda_{-,3}^T(v_{1,2}^-) \lesssim_{\sigma_L} \|u_0^L\|_{H^s(\mathbb{R}^-)}, \qquad 0 \le s < 1. \tag{5.41}$$

To estimate $w_{1,1}^-(x,t)$ and $w_{1,2}^-(x,t)$, we again apply Minkowski's integral inequality to obtain

$$\Lambda_{-,3}^T(w_1^-) \le \Lambda_{-,3}^T(w_{1,1}^-) + \Lambda_{-,3}^T(w_{1,2}^-) \lesssim_{\sigma_L} \int_0^T \|f_L(\cdot,\tau)\|_{H^s(\mathbb{R}^-)}d\tau, \qquad 0 \le s < 1. \tag{5.42}$$

It remains to estimate the functions $H_0^-(x,t)$ and $H_1^-(x,t)$. Using the same change of variables and Lemma 3.4, we obtain

$$\Lambda_{-,3}^T(H_{0,1}^-) + \Lambda_{-,3}^T(H_{0,2}^-) \lesssim_{\sigma_L} \|\mu^*\|_{H^{\frac{s+1}{3}}(\mathbb{R})} \lesssim \|\mu_0\|_{H^{\frac{s+1}{3}}((0,T))}, \qquad \frac{1}{2} < s < 1. \tag{5.43}$$

To estimate the functions $H^-_{1,1}(x,t)$ and $H^-_{1,2}(x,t)$, we apply the same argument in (5.40) and (5.41) to obtain

$$\Lambda^T_{-,3}(H^-_{1,1}) + \Lambda^T_{-,3}(H^-_{1,2}) \lesssim_{\sigma_L} \|\mu_1\|_{H^{\frac{s}{3}}((0,T))}, \qquad 0 \le s < 1. \tag{5.44}$$

□

Theorem 5.4. *Let $3/4 \le s < 1$. Suppose that $u_0^L \in H^s(\mathbb{R}^-)$, $u_0^R \in H^s(\mathbb{R}^+)$, $\mu_0 \in H^{(s+1)/3}((0,T))$, and $\mu_1 \in H^{s/3}((0,T))$. Then, the following estimate holds:*

$$\Lambda^T_{-,4}(v^L) + \Lambda^T_{+,4}(v^R) \lesssim_{s,\sigma_L,\sigma_R} \|(u_0,\mu_0,\mu_1)\|_D + \int_0^T \left(\|f_L(\cdot,t)\|_{H^s(\mathbb{R}^-)} + \|f_R(\cdot,t)\|_{H^s(\mathbb{R}^+)}\right)dt, \tag{5.45}$$

where $v^L(x,t) = S_L[u_0,\mu_0,\mu_1;f]$ and $v^R(x,t) = S_R[u_0,\mu_0,\mu_1;f]$ are the solutions defined in (1.17).

Proof. To establish the estimate in (5.45), it suffices to prove the following inequality:

$$\Lambda^T_{-,4}(\mathcal{U}_L + \mathcal{F}_L + v_1^- + w_1^- + H_0^- + H_1^-) \lesssim_{s,\sigma_L} \|(u_0,\mu_0,\mu_1)\|_D + \int_0^T \|f_L(\cdot,t)\|_{H^s(\mathbb{R}^-)}dt. \tag{5.46}$$

The proof for the others is analogous.
We now consider the estimate of $\mathcal{U}_L(x,t)$. Note that

$$\partial_x S(t)U_0^L(x) = -D_x^{\frac{1}{4}} S(t)\left(D_x^{\frac{3}{4}}\mathcal{H}(U_0^L)\right)(x), \tag{5.47}$$

where $\mathcal{H}(U_0^L)(x)$ denotes the Hilbert transform, defined by

$$\mathcal{H}(U_0^L)(x) := \frac{1}{\pi}P.V.\int_{\mathbb{R}} \frac{U_0^L(y)}{x-y}\,dy, \quad x \in \mathbb{R}. \tag{5.48}$$

Its Fourier transform is given by $\widehat{\mathcal{H}(U_0^L)}(k) = -i\,\mathrm{sgn}(k)\widehat{U_0^L}(k)$. We recall the following result from [10], which provides an estimate for $\partial_x S(t)U_0^L(x)$.

Theorem 5.5 (Theorem 2.4 (2.5) in [10]). *For any $(\theta,\beta) \in [0,1]\times[0,1/2]$, $q = 6/(\theta(\beta+1))$ and $p = 2/(1-\theta)$,*

$$\left(\int_{\mathbb{R}} \left\|D_x^{\frac{\theta\beta}{2}} S(t)v_0\right\|^q_{L^p(\mathbb{R})}\right)^{\frac{1}{q}} \lesssim \|v_0\|_{L^2(\mathbb{R})}, \tag{5.49}$$

where $S(t)v_0(x)$ is defined in (5.35).

Although $S(t)v_0(x)$ in this estimate differs from our case (5.33) by a constant factor σ_L^3, this does not affect the estimate. Applying the estimate in Theorem 5.5 with $p = \infty$, $q = 4$ and $3/4 \le s < 1$, we obtain

$$\Lambda^T_{-,4}(\mathcal{U}_L) \le \left(\int_{\mathbb{R}} \left\|D_x^{\frac{1}{4}} S(t)\left(D_x^{\frac{3}{4}}\mathcal{H}(U_0^L)\right)\right\|^4_{L^\infty(\mathbb{R})} dt\right)^{\frac{1}{4}} \lesssim \left\|D_x^{\frac{3}{4}}\mathcal{H}(U_0^L)\right\|_{L^2(\mathbb{R})} \le \|U_0^L\|_{H^s(\mathbb{R})} \lesssim \|u_0^L\|_{H^s(\mathbb{R}^-)}. \tag{5.50}$$

Next, we estimate $\mathcal{F}_L(x,t)$. Apply Minkowski's integral inequality and the same estimate as in (5.50), we find

$$\Lambda^T_{-,4}(\mathcal{F}_L) \lesssim \int_0^T \|f_L(\cdot,t)\|_{H^s(\mathbb{R}^-)}dt, \qquad 3/4 \le s < 1. \tag{5.51}$$

To establish the remaining terms, we use the following Strichartz-type estimate on the half-line, which is proved in [22].

Lemma 5.6 (Lemma 4.1 in [22]). *For $\zeta \in \mathbb{C}$, we define*

$$\triangle_x^r \mathcal{R}(x,t) := \frac{1}{2\pi}\int_0^\infty e^{i\zeta kx + ik^3 t} k^r \widehat{R}(k)\,dk, \qquad r \in \mathbb{R},\ x \geq 0,\ t \in [0,T] \tag{5.52a}$$

and

$$R(x) := \frac{1}{2\pi}\int_{\mathbb{R}} e^{ikx}\widehat{R}(k)\,dk, \qquad x \geq 0. \tag{5.52b}$$

Given $\widehat{R} \in L^2(\mathbb{R})$, we can define $R \in L^2(\mathbb{R}^+)$ by (5.52b) and $\triangle_x^r \mathcal{R}(x,t)$ by (5.52a). Then, for any $(\theta,\beta) \in [0,1]\times[0,1/2]$ and any $T > 0$, there exists a constant such that

$$\left(\int_0^T \left\| \triangle_x^{\frac{\theta\beta}{2}} \mathcal{R}(\cdot,t)\right\|_{L^p(\mathbb{R}^+)}^q dt\right)^{\frac{1}{q}} \lesssim \|R\|_{L^2(\mathbb{R}^+)} \tag{5.53}$$

with $q = 6/(\theta(\beta+1))$ and $p = 2/(1-\theta)$.

We shall apply this lemma to each of the functions $v_1^-(x,t)$, $w_1^-(x,t)$, $H_0^-(x,t)$, and $H_1^-(x,t)$ separately, with the only difference lying in the specific definition of $\widehat{R}(k)$ in the expression $\triangle_x^r \mathcal{R}(x,t)$. For the function $v_{1,1}^-(x,t)$, we find

$$\partial_x v_{1,1}^-(x,t) = \int_0^\infty e^{-i\frac{\alpha}{\sigma_L}kx + ik^3 t} k^{\frac{1}{4}}\left(\left(-i\frac{\alpha}{\sigma_L}k^{\frac{3}{4}}\right)\mathfrak{v}_{1,1}(k)\right)dk. \tag{5.54}$$

By applying the estimate (5.53) with $p = \infty$ and $q = 4$, we obtain that for $s \geq 3/4$,

$$\Lambda_{-,4}^T(v_{1,1}^-) \lesssim \left(\int_{-\infty}^0 |k|^{\frac{3}{2}}|\mathfrak{v}_{1,1}(k)|^2\,dk\right)^{\frac{1}{2}} \lesssim_{\sigma_L} \|u_0^L\|_{H^s(\mathbb{R}^-)}. \tag{5.55}$$

An analogous argument applied to $v_{1,2}^-(x,t)$ yields

$$\Lambda_{-,4}^T(v_1^-) \lesssim_{\sigma_L} \|u_0^L\|_{H^s(\mathbb{R}^-)}. \tag{5.56}$$

For the function $w_{1,1}^-(x,t)$ and $w_{1,2}^-(x,t)$, we have

$$\partial_x w_{1,1}^-(x,t) = \int_0^t\int_0^\infty e^{-i\frac{\alpha}{\sigma_L}kx + ik^3(t-t')} k^{\frac{1}{4}}\left(\left(-i\frac{\alpha}{\sigma_L}k^{\frac{3}{4}}\right)\mathfrak{F}_{1,1}(k,t')\right)dk\,dt'. \tag{5.57a}$$

$$\partial_x w_{1,2}^-(x,t) = \int_0^t\int_0^\infty e^{i\frac{\tilde{\alpha}}{\sigma_L}kx - ik^3(t-t')} k^{\frac{1}{4}}\left(\left(i\frac{\tilde{\alpha}}{\sigma_L}k^{\frac{3}{4}}\right)\mathfrak{F}_{1,2}(k,t')\right)dk\,dt'. \tag{5.57b}$$

By applying Minkowski's integral inequality and the estimate (5.53) with $p = \infty$ and $q = 4$, we obtain that for $3/4 \leq s < 1$,

$$\Lambda_{-,4}^T(w_1^-) \leq \Lambda_{-,4}^T(w_{1,1}^-) + \Lambda_{-,4}^T(w_{1,2}^-) \lesssim_{\sigma_L} \int_0^T \|f_L(\cdot,t')\|_{H^s(\mathbb{R}^-)}dt'. \tag{5.58}$$

Finally, for the functions $H_0^-(x,t)$ and $H_1^-(x,t)$, we have

$$\partial_x H_{0,1}^-(x,t) = \int_0^\infty e^{-i\frac{\alpha}{\sigma_L}kx + ik^3 t} k^{\frac{1}{4}}\left(\left(-i\frac{\alpha}{\sigma_L}k^{\frac{3}{4}}\right)k^2 M_0(-ik^3,T)\right)dk, \tag{5.59a}$$

$$\partial_x H_{0,2}^-(x,t) = \int_0^\infty e^{i\frac{\tilde{\alpha}}{\sigma_L}kx + ik^3 t} k^{\frac{1}{4}}\left(\left(i\frac{\tilde{\alpha}}{\sigma_L}k^{\frac{3}{4}}\right)k^2 M_0(-ik^3,T)\right)dk, \tag{5.59b}$$

and

$$\partial_x H^-_{1,1}(x,t) = \int_0^\infty e^{-i\frac{\alpha}{\sigma_L}kx+ik^3t} k^{\frac{1}{4}} \left(\left(-i\frac{\alpha}{\sigma_L} k^{\frac{3}{4}} \right) \left(-\alpha^2 ikM_1(-ik^3,T) \right) \right) dk, \tag{5.59c}$$

$$\partial_x H^-_{1,2}(x,t) = \int_0^\infty e^{i\frac{\tilde{\alpha}}{\sigma_L}kx+ik^3t} k^{\frac{1}{4}} \left(\left(i\frac{\tilde{\alpha}}{\sigma_L} k^{\frac{3}{4}} \right) \left(-\alpha^2 ikM_1(-ik^3,T) \right) \right) dk. \tag{5.59d}$$

By applying the estimate (5.53) with $p = \infty$ and $q = 4$ again, we conclude that for $3/4 \le s < 1$,

$$\Lambda^T_{-,4}(H^-_0) \le \Lambda^T_{-,4}(H^-_{0,1}) + \Lambda^T_{-,4}(H^-_{0,2}) \lesssim_{\sigma_L} \|\mu^*\|_{H^{\frac{s+1}{3}}(\mathbb{R})} \lesssim \|\mu_0\|_{H^{\frac{s+1}{3}}((0,T))}, \tag{5.60a}$$

and

$$\Lambda^T_{-,4}(H^-_1) \le \Lambda^T_{-,4}(H^-_{1,1}) + \Lambda^T_{-,4}(H^-_{1,2}) \lesssim_{\sigma_L} \|\mu_1\|_{H^{\frac{s}{3}}((0,T))}. \tag{5.60b}$$

□

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Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

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